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Graph Traffic Forecasting under Distribution Shift

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1 Introduction

Accurate short-term knowledge of traffic conditions is essential for understanding and managing the dynamics of urban road networks. However, traffic evolution is difficult to predict because it results from the interaction of two strongly coupled components. On the one hand, traffic at a given location follows temporal patterns determined by its recent history; on the other hand, changes occurring on one road segment can propagate through the network and affect other connected locations. A forecasting model should therefore be able to exploit both temporal dependencies and the spatial structure of the road network.

Accurate point forecasts alone, however, are not sufficient in practical settings. Traffic conditions may change rapidly, particularly during congestion events, making some predictions substantially more uncertain than others. A forecasting system should therefore provide not only an estimate of future traffic conditions, but also an indication of the uncertainty associated with that estimate. Moreover, this uncertainty should remain meaningful when the model encounters conditions that differ from those observed during training.

This work investigates these aspects using the METR-LA traffic-sensor network. Starting from a simple persistence baseline, a temporal forecasting model based on Gated Recurrent Units (GRUs) is first considered to model the temporal evolution of each sensor independently. Spatial information is then introduced through a graph-temporal architecture combining diffusion-style message passing over the road-sensor graph with recurrent temporal modeling. Both learned models produce probabilistic forecasts through conditional quantile estimation, allowing point accuracy and predictive uncertainty to be evaluated jointly.

Beyond aggregate forecasting performance, particular attention is devoted to understanding the contribution of the graph and the robustness of the probabilistic predictions. The graph-temporal model is analyzed at the sensor level, under different graph configurations and diffusion depths, and under a distribution shift defined by unusually high network-wide congestion. Finally, mitigation strategies are investigated to determine whether the degradation in forecasting accuracy and uncertainty reliability can be reduced when forecasting conditions become more challenging.

1.1 Objectives and Research Questions

The main objective of this work is to design and evaluate a compact, probabilistic graph-temporal forecasting pipeline while investigating the interaction between predictive accuracy, spatial information, uncertainty calibration, and robustness under distribution shift.

More specifically, the experimental analysis is organized around the following research questions:

- 1. Does graph message passing improve traffic forecasting compared with temporal modeling alone?**

This question is investigated by comparing a persistence baseline, a temporal GRU without spatial information exchange, and a graph-temporal model that propagates information between connected traffic sensors.

- 2. How does the graph representation affect forecasting performance?**

The contribution of the graph is examined both at the individual-sensor level and through controlled architectural experiments. In particular, the analysis investigates where graph message passing helps or hurts, whether extending the spatial diffusion depth provides additional benefits, and whether the original weighted graph contains useful information beyond connectivity alone.

- 3. How reliable are the probabilistic forecasts under normal and shifted traffic conditions?**

Forecast uncertainty is evaluated through quantile-based predictive intervals, considering both empirical coverage and interval width. A predefined high-congestion stress test is then used to investigate how point accuracy and uncertainty calibration change when the model is evaluated under more challenging traffic conditions.

- 4. Can the calibration degradation observed under distribution shift be mitigated?**

A post-hoc Conformalized Quantile Regression calibration procedure and a congestion-aware training strategy are evaluated as complementary mitigation approaches. The former directly targets predictive-interval calibration, whereas the latter modifies the training objective to assign greater importance to rare high-congestion targets.

Together, these questions shift the analysis beyond average forecasting accuracy and toward a more complete assessment of when spatial information is useful, when the model becomes uncertain, and whether its uncertainty estimates remain reliable under distribution shift.

2 Dataset and Preprocessing

2.1 METR-LA Dataset

The experiments were conducted on the METR-LA dataset, which contains traffic-speed measurements collected from 207 loop detectors located on the Los Angeles County road network. Measurements are sampled every five minutes, resulting in 34272 timestamps covering the period from March 1 to June 27, 2012. Each sensor is associated with a single feature representing the observed traffic speed.

The dataset can therefore be represented as a multivariate time series $\mathbf{X} \in \mathbb{R}^{T \times N}$, where $T = 34272$ denotes the number of timestamps and $N = 207$ the number of traffic sensors. The spatial relationships between sensors are represented by the road-network graph released with the public DCRNN implementation for METR-LA [1]. Its adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ is directed and weighted and is later used by the graph-temporal model for spatial message passing. A detailed description of the graph construction and normalization is provided in Section 3.4.

2.2 Exploratory Data Analysis

Before defining the preprocessing pipeline, an exploratory analysis was performed to investigate the temporal structure, data quality, and variability of the traffic measurements. The temporal index was found to be regularly spaced, with no duplicated timestamps and no explicit NaN or infinite values in the original data.

Particular attention was devoted to zero-valued measurements. Approximately 8.11% of all observations are equal to zero, and zero values occur for all 207 sensors, although with considerably different frequencies across nodes. More importantly, several timestamps contain simultaneous zero measurements across the entire sensor network. The longest consecutive sequence consists of 381 timestamps, corresponding to approximately 31 hours and 45 minutes during which no sensor reports a positive traffic speed.

The temporal concentration of these measurements suggests that a substantial fraction of the zero values represents unavailable or invalid sensor readings rather than actual zero traffic speed. This behavior is illustrated in Figure 1. When zero values are included in the network-wide average, several abrupt drops appear in correspondence with periods in which a large number of sensors simultaneously report zero measurements. After excluding zero-valued observations, these artificial drops disappear and intervals in which the entire network is unavailable appear as gaps in the time series.

This provides additional evidence that zero measurements should not be interpreted directly as actual traffic speeds. Based on this analysis, zero-valued measurements were therefore treated as missing observations in the subsequent preprocessing pipeline.

The valid traffic measurements also exhibit strong temporal structure. Aggregating traffic speeds by hour of the day reveals a clear daily seasonality, with average speeds decreasing during typical congestion periods and increasing again during less congested hours. However, aggregating all days together partially hides differences between working days and weekends.

Figure 2 therefore compares the hourly traffic-speed profiles obtained separately for weekdays and weekends. Weekdays exhibit two pronounced reductions in average speed, occurring around 8:00 and 17:00 and corresponding to the typical morning and afternoon commuting periods. In contrast, weekends show a smoother daily profile, with generally higher speeds during these hours. The figure therefore reveals a weekly component in addition to the daily seasonality.

These observations confirm that the traffic series contains meaningful temporal dependencies at multiple time scales and motivate the use of a temporal forecasting model in the subsequent experiments.

Finally, the distribution of traffic speeds was examined at the individual sensor level. Considerable heterogeneity was observed across the network, since the mean traffic speed ranges from approximately 30.89 to 67.73 across sensors, while the corresponding temporal standard deviations range from approximately 1.99 to 23.94. Therefore, different sensors exhibit distinct traffic levels and degrees of variability.

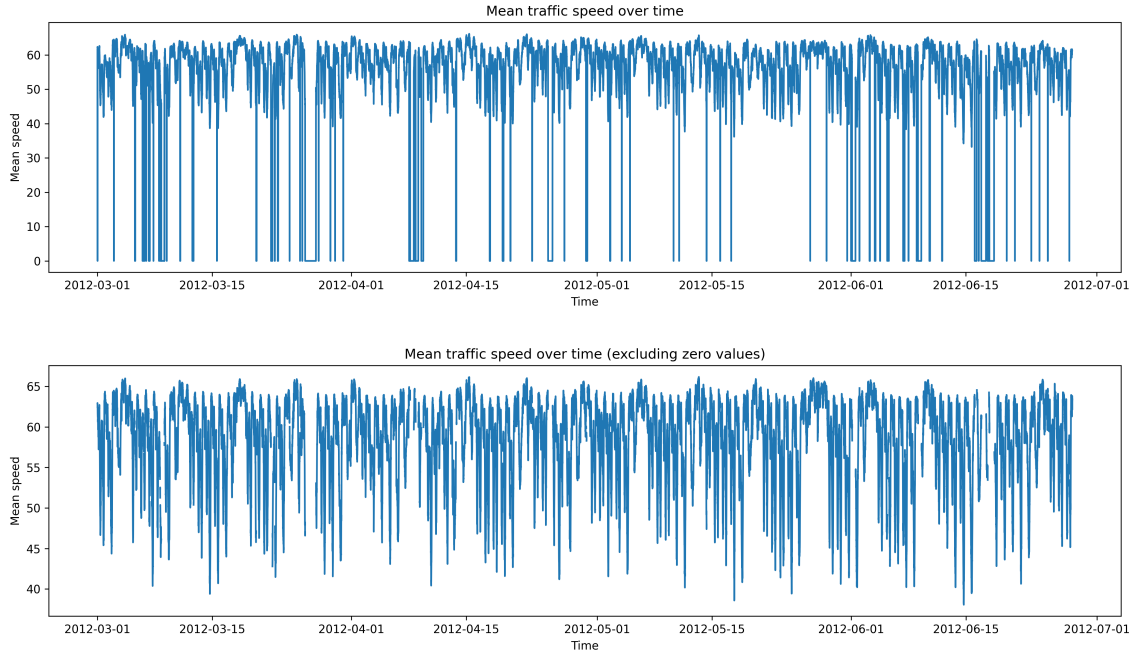


Figure 1: Mean traffic speed over time before (top) and after (bottom) excluding zero values. Removing zero-valued measurements eliminates the abrupt drops to zero while preserving the temporal traffic patterns.

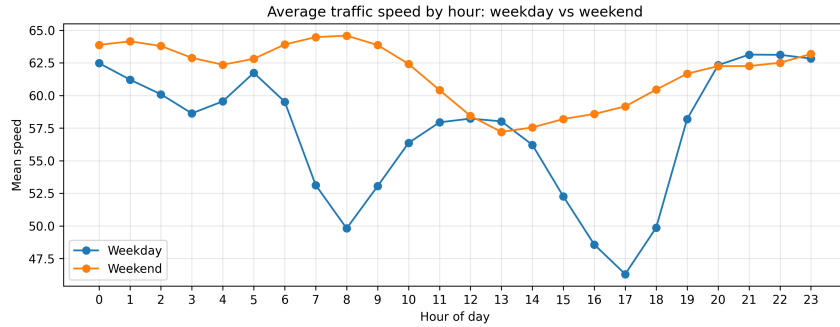


Figure 2: Average traffic-speed profile by hour of the day, computed separately for weekdays and weekends. The different profiles highlight both daily and weekly temporal patterns in the METR-LA measurements.

This heterogeneity is particularly relevant for preprocessing, since applying a single global normalization would ignore the different operating ranges of individual sensors. For this reason, the normalization strategy described in Section 2.4 estimates separate statistics for each sensor. Overall, the exploratory analysis identifies two main properties that guide the preprocessing and modeling choices adopted in this work: a non-negligible amount of unavailable measurements encoded as zeros, and strong but heterogeneous temporal traffic patterns across the sensor network.

2.3 Leakage-Free Data Partitioning

To preserve the temporal nature of the forecasting problem and prevent information leakage, the original time series was partitioned chronologically, without shuffling, into training, validation, and test sets using a 70%-10%-20% split. Importantly, the partition was performed on the raw time series before normalization and temporal window construction.

The resulting partitions are summarized in Table 1. The training set contains 23990 timestamps, followed by 3427 validation timestamps and 6855 test timestamps. This ordering ensures that validation and test observations always occur later in time than the observations used for model training.

All data-dependent preprocessing operations were subsequently fitted using the training partition only. In particular, the per-sensor normalization statistics were estimated exclusively from training observations and then reused unchanged for validation and test data.

Table 1: Chronological partition of the original METR-LA time series before preprocessing and temporal window construction.

Split	Proportion	Timestamps	Time period
Training	70%	23990	Mar. 1 00:00 – May 23 07:05
Validation	10%	3427	May 23 07:10 – Jun. 4 04:40
Test	20%	6855	Jun. 4 04:45 – Jun. 27 23:55

Temporal windows were also constructed independently within each partition. Consequently, an input–target pair can never cross a split boundary. For example, a training input window cannot use observations from the validation period as future targets. This procedure preserves the chronological separation between model development and final evaluation.

2.4 Normalization and Temporal Windowing

As discussed in Section 2.2, traffic-speed distributions vary substantially across sensors. For this reason, the data were standardized independently for each sensor rather than using global statistics shared across the complete network.

For each sensor n , the mean μ_n and standard deviation σ_n were estimated exclusively from the valid observations in the training partition. Each available traffic-speed observation $x_{t,n}$ was then transformed as

$$\tilde{x}_{t,n} = \frac{x_{t,n} - \mu_n}{\sigma_n}. \quad (1)$$

The same training statistics were subsequently applied without modification to the validation and test partitions. This prevents information from future partitions from influencing the preprocessing procedure.

After preprocessing and normalization, each chronological partition was independently transformed into input–target pairs using a sliding-window procedure. Given an input-history length $H = 12$ and a forecasting window $F = 12$, a sample generated at time t consists of the previous 12 traffic observations and the following 12 observations to be predicted.

More formally, the input and target windows are defined as

$$\mathbf{X}_t = [\tilde{\mathbf{x}}_{t-H+1}, \dots, \tilde{\mathbf{x}}_t] \in \mathbb{R}^{H \times N}, \quad (2)$$

and

$$\mathbf{Y}_t = [\tilde{\mathbf{x}}_{t+1}, \dots, \tilde{\mathbf{x}}_{t+F}] \in \mathbb{R}^{F \times N}, \quad (3)$$

where $N = 207$ is the number of sensors. Since measurements are sampled every five minutes, both $H = 12$ and $F = 12$ correspond to one hour of traffic observations.

Although the models jointly predict all $F = 12$ future steps, the main evaluation is performed at three representative forecasting horizons: $t + 3$, $t + 6$, and $t + 12$, corresponding respectively to 15, 30, and 60 minutes into the future. This allows performance to be analyzed as the forecasting task becomes progressively more difficult with increasing prediction horizon.

The windowing procedure was applied jointly to the traffic values and their availability masks. Consequently, each input window is associated with an input mask indicating which historical observations are available, while each target window is associated with a target mask identifying the entries that can contribute to the training loss and evaluation metrics.

Partially observed windows are therefore retained whenever useful information is available, rather than being discarded entirely because of a small number of missing sensor measurements. As described in Section 2.5, unavailable target entries never contribute to model optimization or performance evaluation.

The resulting preprocessing pipeline therefore preserves the chronological structure of the forecasting problem, estimates all data-dependent transformations exclusively from the training period, and explicitly tracks unavailable observations through masks. The same processed input-target windows are used by all forecasting approaches described in the following section, ensuring that differences in performance are attributable to the forecasting models rather than to different data-processing procedures.

Table 2: Number of temporal samples retained after missing-data processing and filtering.

Split	Final samples
Training	22817
Validation	3240
Test	6224

2.5 Missing-Value Handling

Based on the exploratory analysis in Section 2.2, zero-valued traffic measurements were interpreted as unavailable observations and replaced with missing values before further processing. After normalization and temporal window construction, missing values were handled differently in the input histories X and forecasting targets Y , since imputing a target would create artificial ground-truth values and could bias both training and evaluation.

Input histories. Missing observations in each input window were handled through a limited interpolation procedure. For each sensor independently, only short internal gaps of at most three consecutive timestamps were interpolated, and only when valid observations were available on both sides of the gap. Longer gaps and missing values at the boundaries of an input window were left unchanged.

This conservative strategy reduces isolated sensor dropouts while avoiding the reconstruction of long unavailable periods from potentially unrelated observations. Before interpolation, missing values accounted for approximately 7.13%, 7.00%, and 12.13% of the input values in the training, validation, and test windows, respectively. After short-gap interpolation, these proportions decreased to approximately 6.85%, 6.83%, and 11.92%.

Therefore, interpolation only moderately reduces the overall amount of missing data. This is expected because the procedure deliberately targets short internal gaps while preserving longer periods of sensor unavailability.

Input windows for which all sensors were unavailable throughout the complete history were removed, since they contain no temporal information from which a forecast can be produced. In contrast, a window was retained when only a subset of sensors was fully unavailable, because observations from the remaining sensors may still provide useful information, particularly for the graph-temporal model.

After this filtering step, residual missing input values were replaced by zero in the standardized domain. Since normalization is performed separately for each sensor using training statistics, this value corresponds to the training mean of that sensor rather than to a physical traffic speed of zero. An input availability mask was retained to distinguish observed from missing measurements.

Forecasting targets. A stricter policy was adopted for the forecasting targets. Target values were never interpolated or semantically imputed, since doing so would introduce synthetic ground truth into the optimization and evaluation procedures. Instead, a binary target-availability mask was constructed and propagated throughout the complete pipeline.

Windows with no valid target observations over the complete forecasting period were removed. Partially observed target windows were retained, while missing target entries were numerically replaced with finite placeholder values only to maintain valid tensor representations. These entries were always excluded from the loss and evaluation metrics through the target mask and therefore never contributed to model optimization or reported performance.

After filtering fully unavailable input and target windows, the final dataset contains 22817 training samples, 3240 validation samples, and 6224 test samples, as reported in Table 2.

Despite the missing periods identified in the raw data, target availability remains high at all evaluated horizons, ranging from approximately 95% to 97% depending on the split and forecast horizon, providing a sufficiently large set of valid observations for model training and evaluation.

3 Forecasting Methodology

3.1 Probabilistic Forecasting Formulation

The forecasting task is formulated probabilistically through conditional quantile estimation. Rather than predicting a single future traffic speed, the neural forecasting models estimate three quantiles

for each sensor and forecasting step,

$$\mathcal{Q} = \{0.1, 0.5, 0.9\}. \quad (4)$$

For an input window $\mathbf{X}_t$, the models therefore produce

$$\hat{\mathbf{Y}}_t \in \mathbb{R}^{F \times N \times |\mathcal{Q}|}, \quad (5)$$

where $F = 12$ is the forecasting window, $N = 207$ is the number of sensors, and $|\mathcal{Q}| = 3$ is the number of predicted quantiles.

The median prediction $\hat{y}^{(0.5)}$ is used as the point forecast when computing deterministic metrics such as MAE and RMSE. The lower and upper quantiles define the predictive interval

$$\left[\hat{y}^{(0.1)}, \hat{y}^{(0.9)} \right], \quad (6)$$

which has a nominal coverage probability of 80%. This formulation makes it possible to evaluate point forecasting accuracy and predictive uncertainty using a single model.

The models are trained using the Pinball Loss. For a target value y , a predicted quantile $\hat{y}^{(q)}$, and a quantile level q , the loss is defined as

$$\mathcal{L}_q \left(y, \hat{y}^{(q)} \right) = \max \left(q \left(y - \hat{y}^{(q)} \right), (q - 1) \left(y - \hat{y}^{(q)} \right) \right). \quad (7)$$

Unlike a symmetric regression loss, the Pinball Loss penalizes underestimation and overestimation differently according to the target quantile. Minimizing it therefore allows the model to estimate different regions of the conditional target distribution rather than only its central value.

Since some target observations are unavailable, the Pinball Loss is computed only over valid target entries. During training, consider a mini-batch of B samples. Let $b \in \{1, \dots, B\}$ denote the sample index, $h \in \{1, \dots, F\}$ the forecast step, and $n \in \{1, \dots, N\}$ the sensor index. The binary mask $m_{b,h,n} \in \{0, 1\}$ indicates whether the corresponding target value is available.

For each mini-batch, the training loss is obtained by averaging the Pinball Loss over all valid target entries and over all predicted quantiles:

$$\mathcal{L}_{\text{masked}} = \frac{\sum_{b=1}^B \sum_{h=1}^F \sum_{n=1}^N m_{b,h,n} \sum_{q \in \mathcal{Q}} \mathcal{L}_q \left(y_{b,h,n}, \hat{y}_{b,h,n}^{(q)} \right)}{|\mathcal{Q}| \sum_{b=1}^B \sum_{h=1}^F \sum_{n=1}^N m_{b,h,n}}. \quad (8)$$

Consequently, unavailable target values do not contribute to the loss, while every valid target contributes once for each quantile in $\mathcal{Q}$. This is consistent with the missing-data strategy described in Section 2.5.

3.2 Persistence Baseline

A persistence model is used as the simplest forecasting baseline. For each sensor, the value at the final time step of the input window is repeated across the complete forecasting horizon. The prediction for sensor n at any future step h is therefore

$$\hat{y}_{t+h,n} = x_{t,n}, \quad h = 1, \dots, F. \quad (9)$$

The persistence baseline contains no trainable parameters and does not explicitly model either temporal dynamics or spatial interactions. It therefore provides a simple reference for determining whether the learned models extract useful predictive information from the historical traffic sequence.

3.3 Temporal GRU

The first learned forecasting model is a temporal Gated Recurrent Unit (GRU), which serves as the no-graph baseline. Each sensor is processed independently using its own sequence of historical traffic observations, while the GRU parameters are shared across all sensors.

For a given sensor n , the input sequence

$$[\tilde{x}_{t-H+1,n}, \dots, \tilde{x}_{t,n}] \quad (10)$$

is processed by the same recurrent network. No information from other sensors is exchanged during the forward pass. The model therefore provides a controlled no-graph baseline against which the contribution of spatial message passing can be evaluated.

The selected temporal architecture consists of a two-layer GRU with hidden dimension 64. The final hidden representation of each sensor is passed to a linear output layer that jointly predicts all $F = 12$ future steps and all three quantiles. For each sensor, the output layer therefore produces

$$F \times |\mathcal{Q}| = 12 \times 3 = 36 \quad (11)$$

values, which are reshaped into the corresponding forecast-step and quantile dimensions.

The complete model contains 40164 trainable parameters. Because the same network is applied to every sensor, the parameter count does not scale with the number of nodes in the traffic network.

3.4 Graph Representation and Diffusion-Style Message Passing

Spatial dependencies are introduced through the sensor graph provided with METR-LA [1]. Each traffic sensor corresponds to a graph node, while the adjacency matrix

$$\mathbf{A} \in \mathbb{R}^{N \times N} \quad (12)$$

encodes directed, weighted connections between sensors. Throughout this work, the convention is that A_{ij} represents the weight assigned at node i to the message received from node j . Therefore, row i contains the weights used to aggregate information into node i .

The provided adjacency matrix already contains self-connections. Therefore, no additional identity matrix is added to $\mathbf{A}$ before normalization. This avoids introducing artificial self-loops or modifying the relative weights of the graph supplied with the dataset.

Before message passing, the adjacency matrix is row-normalized as

$$\mathbf{P} = \mathbf{D}^{-1} \mathbf{A}, \quad (13)$$

where $\mathbf{D}$ is the diagonal matrix of row sums,

$$D_{ii} = \sum_j A_{ij}. \quad (14)$$

Under the adopted convention, the one-step propagated representation of node i is therefore

$$(\mathbf{P}\mathbf{X})_i = \sum_j P_{ij} \mathbf{X}_j, \quad (15)$$

so that node i receives a weighted combination of the representations of the nodes j connected to it.

Spatial information is incorporated through a diffusion-style message-passing layer, conceptually inspired by diffusion-based graph modeling for traffic forecasting [1]. This formulation implements a compact unidirectional multi-hop propagation operator rather than the bidirectional diffusion convolution of DCRNN. Only powers of the row-normalized propagation matrix $\mathbf{P}$ are considered, with a separate learnable transformation $\mathbf{W}_k$ for each propagation depth. Given node features $\mathbf{X}$, the layer computes

$$\mathbf{H} = \sum_{k=0}^K \mathbf{P}^k \mathbf{X} \mathbf{W}_k + \mathbf{b}, \quad (16)$$

where K controls the maximum diffusion depth and each $\mathbf{W}_k$ is a learnable projection associated with a different propagation step. The $k = 0$ term, for which $\mathbf{P}^0 = \mathbf{I}$, directly preserves the node's own representation, while the terms with $k \geq 1$ incorporate information propagated through the graph.

For the main configuration, $K = 2$, yielding

$$\mathbf{H} = \mathbf{X} \mathbf{W}_0 + \mathbf{P} \mathbf{X} \mathbf{W}_1 + \mathbf{P}^2 \mathbf{X} \mathbf{W}_2 + \mathbf{b}. \quad (17)$$

The first term provides an explicit transformation of each node's own representation, while the remaining terms incorporate information propagated over one and two graph transitions. Since the diagonal entries already present in $\mathbf{A}$ are preserved by the row normalization, the one-step and two-step diffusion terms may themselves also contain self-information.

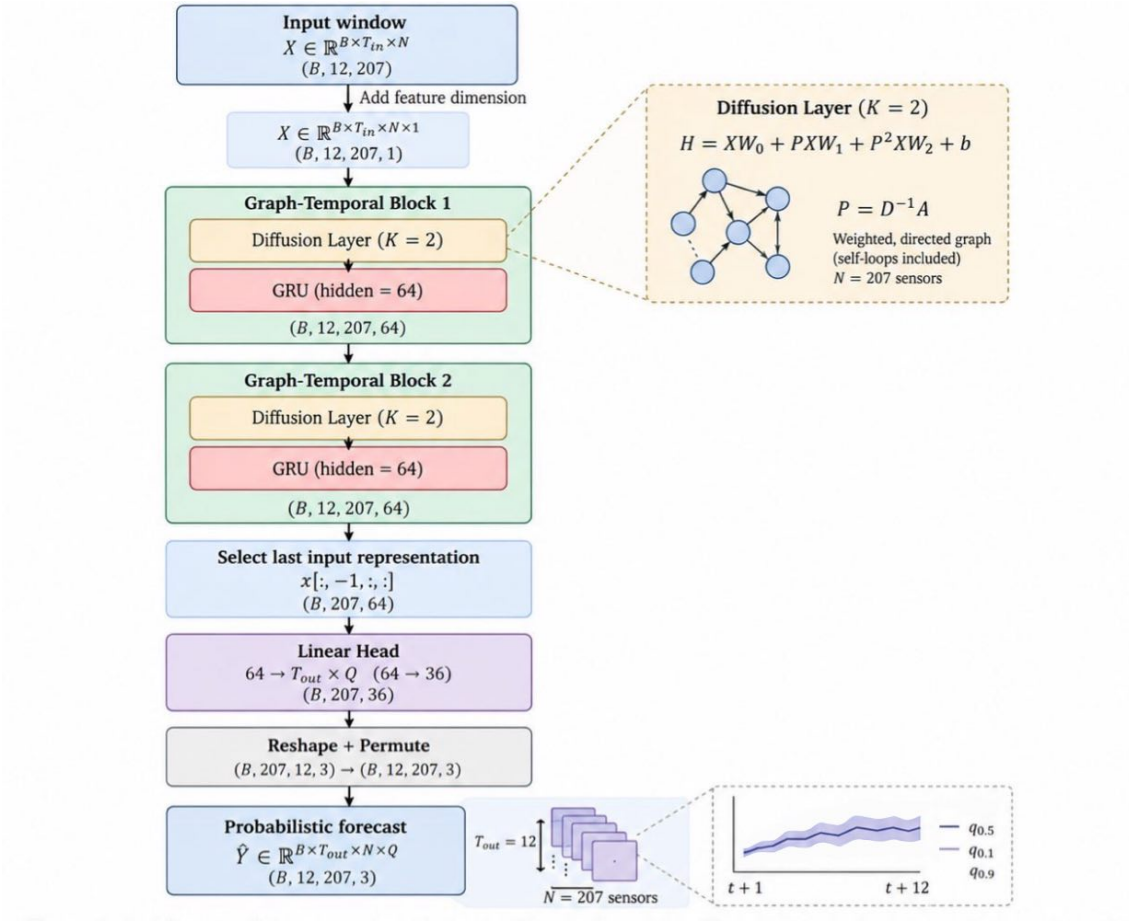


Figure 3: Architecture of the proposed graph-temporal forecasting model.

3.5 Graph-Temporal Architecture

The main forecasting model combines the diffusion-style message-passing operation described in Section 3.4 with recurrent temporal modeling. The architecture is composed of two consecutive graph-temporal blocks. Each block first propagates information between sensors through diffusion-style message passing and then applies a GRU to model the temporal evolution of the resulting node representations (Figure 3).

Given an input window, diffusion is applied independently at each temporal step, allowing every sensor representation to incorporate information from its graph neighborhood. The resulting sequence of spatial representations is then processed by a GRU independently for each node, with recurrent parameters shared across sensors. The first block preserves the complete temporal sequence so that it can be processed by the second diffusion-GRU block.

The final temporal representation is passed to a linear probabilistic head that directly predicts the 12 future steps for the three quantile levels defined in Section 3.1. As in the Temporal GRU, the complete forecasting window is generated in a single forward pass rather than recursively feeding previous predictions back into the model.

The selected graph-temporal model uses a graph hidden dimension of 64, a GRU hidden dimension of 64, one GRU layer per graph-temporal block, two graph-temporal blocks, and a diffusion depth of $K = 2$. The complete architecture contains 64868 trainable parameters.

4 Experimental Setup

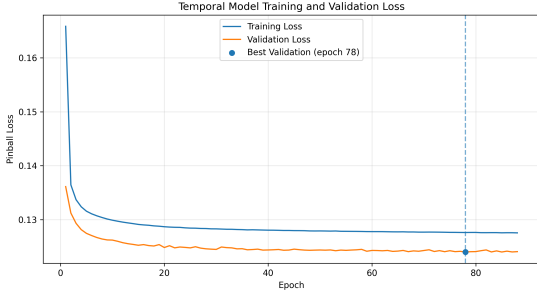
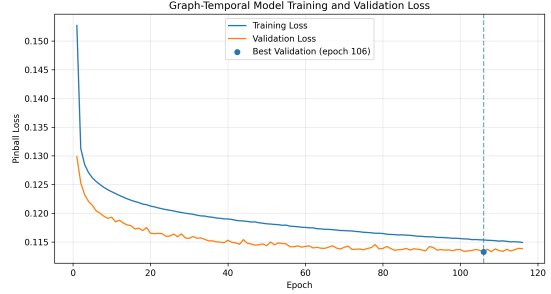
4.1 Training Protocol

Both the Temporal GRU and the Graph-Temporal model were trained using the masked Pinball Loss defined in Equation 8. Model parameters were optimized with Adam using mini-batches of 64 samples.

Training was performed for a maximum of 200 epochs with early stopping based on the validation Pinball Loss and a patience of 10 epochs. Whenever the validation loss improved, the corresponding

Table 3: Hyperparameter search spaces and selected configurations for the two neural forecasting models.

Model	Hyperparameter	Values
Temporal GRU	Hidden dimension	$\{32, \mathbf{64}\}$
	GRU layers	$\{1, \mathbf{2}\}$
	Learning rate	$\{5 \times 10^{-4}, 10^{-3}\}$
Graph-Temporal	Graph hidden dimension	$\{32, \mathbf{64}\}$
	GRU hidden dimension	$\mathbf{64}$
	GRU layers per block	$\{1, \mathbf{2}\}$
	Learning rate	$\{5 \times 10^{-4}, 10^{-3}\}$
	Diffusion depth K	$\mathbf{2}$
	Graph-temporal blocks	$\mathbf{2}$

**(a)** Temporal model.**(b)** Graph-Temporal model.**Figure 4:** Training and validation Pinball Loss for the selected model configurations.

model checkpoint was retained. Training was terminated if no further improvement was observed for 10 consecutive epochs, and the checkpoint associated with the lowest validation loss was used for subsequent evaluation.

Both neural models use direct multi-horizon forecasting: all 12 future time steps and the three target quantiles are predicted simultaneously from the input history.

No test-set information was used for hyperparameter selection, early stopping, or model calibration. The test partition was accessed only after the model configuration had been selected using the training and validation partitions.

4.2 Hyperparameter Selection

Hyperparameters were selected exclusively according to the validation Pinball Loss. Separate grid searches were performed for the Temporal GRU and the Graph-Temporal model, since the two architectures expose different model-specific parameters. The search spaces and selected configurations are summarized in Table 3.

Selected values are highlighted in bold. The diffusion depth and number of graph-temporal blocks were kept fixed during the main hyperparameter search; the effect of changing the diffusion depth is investigated separately in Section 6.2.

For the Temporal model, the selected configuration uses a hidden dimension of 64, two recurrent layers, and a learning rate of 5×10^{-4} . The best checkpoint was obtained at epoch 78, with a validation Pinball Loss of 0.1240.

For the Graph-Temporal model, the selected configuration uses a graph hidden dimension of 64, a GRU hidden dimension of 64, one recurrent layer per graph-temporal block, and a learning rate of 5×10^{-4} . With two blocks and diffusion depth $K = 2$, the best checkpoint was obtained at epoch 106, reaching a validation Pinball Loss of 0.1133.

The learning curves of the selected configurations are shown in Figure 4. For both models, the training loss decreases progressively, while the validation loss eventually reaches a minimum and ceases to improve. The selected checkpoints therefore correspond to the minimum validation loss rather than to the final training epoch, consistently with the early-stopping procedure described in Section 4.1.

Validation Pinball Loss was used as the single model-selection criterion rather than selecting configurations directly according to empirical coverage. This criterion evaluates the quality of the

predicted quantiles without optimizing the model specifically for a single coverage statistic. Coverage and interval width were instead retained as complementary reliability metrics for the final analysis.

4.3 Evaluation Metrics

Performance is evaluated at the three forecasting horizons of 15, 30, and 60 minutes. All metrics are computed only over valid target observations using the corresponding target availability mask. Point forecasting accuracy is evaluated using the median prediction $\hat{y}^{(0.5)}$. The Mean Absolute Error (MAE) is defined as

$$\text{MAE} = \frac{1}{|\Omega|} \sum_{i \in \Omega} |y_i - \hat{y}_i^{(0.5)}|, \quad (18)$$

where Ω denotes the set of valid target entries at the considered forecast horizon. The Root Mean Squared Error (RMSE) is defined as

$$\text{RMSE} = \sqrt{\frac{1}{|\Omega|} \sum_{i \in \Omega} (y_i - \hat{y}_i^{(0.5)})^2}. \quad (19)$$

While MAE measures the average magnitude of the forecasting error, RMSE assigns greater weight to large deviations and is therefore useful for identifying models that occasionally produce particularly inaccurate forecasts.

Probabilistic forecasting quality is evaluated through the Pinball Loss introduced in Equation 7. The loss is reported separately for $q_{0.1}$, $q_{0.5}$, and $q_{0.9}$, allowing the quality of the lower, median, and upper conditional quantile estimates to be examined individually.

The reliability of the predictive intervals is assessed through empirical coverage. For the nominal 80% interval $[\hat{y}^{(0.1)}, \hat{y}^{(0.9)}]$, coverage is computed as

$$\text{Coverage} = \frac{1}{|\Omega|} \sum_{i \in \Omega} \mathbb{I}[\hat{y}_i^{(0.1)} \leq y_i \leq \hat{y}_i^{(0.9)}], \quad (20)$$

where $\mathbb{I}[\cdot]$ is the indicator function. A well-calibrated 80% predictive interval is expected to contain approximately 80% of the observed targets.

Coverage alone is not sufficient to characterize forecast quality, since arbitrarily wide intervals could achieve high coverage. Predictive sharpness is therefore evaluated using the mean interval width,

$$\text{Width} = \frac{1}{|\Omega|} \sum_{i \in \Omega} (\hat{y}_i^{(0.9)} - \hat{y}_i^{(0.1)}). \quad (21)$$

Coverage and interval width are interpreted jointly: reliable forecasts should achieve coverage close to the nominal level while keeping predictive intervals as informative as possible.

Finally, the quantile crossing rate measures the fraction of predictions for which the estimated quantiles violate their expected ordering,

$$\hat{y}^{(0.1)} \leq \hat{y}^{(0.5)} \leq \hat{y}^{(0.9)}. \quad (22)$$

A crossing rate close to zero indicates that the independently predicted quantiles are almost always ordered consistently.

In addition to these scalar metrics, calibration plots are used to compare the nominal quantile levels with their empirical frequencies. This provides a more detailed view of probabilistic calibration than the 80% interval coverage alone.

4.4 Hardware and Reproducibility

All neural models were implemented in PyTorch and trained using an NVIDIA GeForce RTX 4060 Laptop GPU with CUDA acceleration. A fixed random seed of 42 was used throughout the experiments to improve reproducibility.

Model checkpoints corresponding to the best validation Pinball Loss were saved during training and subsequently reloaded for final evaluation. The same preprocessing pipeline, dataset partitions, forecast horizons, and evaluation functions were used across the learned models to ensure a consistent experimental comparison.

Table 4: Point forecasting performance on the test set at the three forecasting horizons. The best result for each metric and horizon is highlighted in bold.

Model	Metric	Forecast horizon		
		15 min	30 min	60 min
Persistence	MAE	3.501	4.199	5.351
	RMSE	6.411	8.023	10.279
Temporal GRU	MAE	3.080	3.735	4.733
	RMSE	6.081	7.548	9.411
Graph-Temporal	MAE	2.901	3.442	4.244
	RMSE	5.511	6.758	8.378

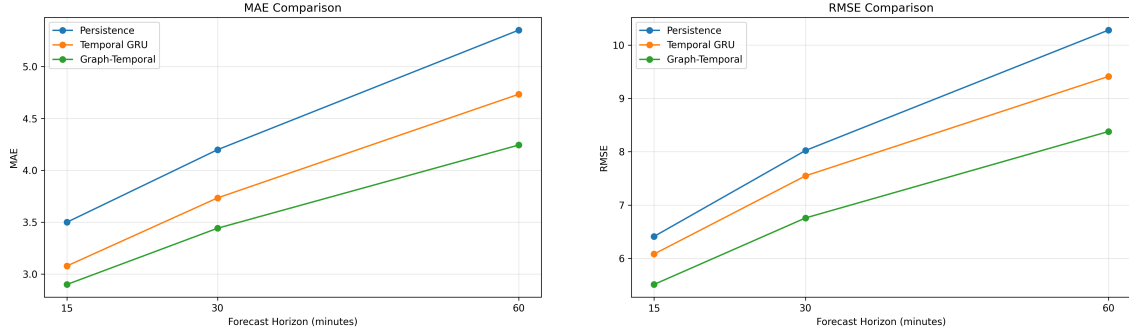


Figure 5: Point forecasting performance of the Persistence, Temporal, and Graph-Temporal models across the three forecasting horizons in terms of MAE (left) and RMSE (right).

5 In-Distribution Forecasting Results

5.1 Point Forecasting Performance

The point forecasting performance of the three models is reported in Table 4. For the two probabilistic neural models, the median prediction $q_{0.5}$ is used as the point forecast. As expected, forecasting accuracy decreases as the prediction horizon increases, with both MAE and RMSE increasing from 15 to 60 minutes for all models.

The Persistence baseline exhibits the largest errors at all forecasting horizons. Introducing temporal modeling through the GRU consistently improves over persistence, showing that the recent history contains temporal patterns that cannot be captured by simply propagating the final input value into the future.

The Graph-Temporal model further reduces both MAE and RMSE at every horizon. Compared with the Temporal model, the MAE decreases from 3.080 to 2.901 at 15 minutes, from 3.735 to 3.442 at 30 minutes, and from 4.733 to 4.244 at 60 minutes. These correspond to relative reductions of approximately 5.8%, 7.8%, and 10.3%, respectively. The corresponding RMSE reductions are approximately 9.4%, 10.5%, and 11.0%.

The improvement introduced by the graph component becomes more pronounced as the forecasting horizon increases. This suggests that spatial information is particularly useful for longer-term predictions, where the future state of a sensor becomes increasingly difficult to estimate from its own temporal history alone. The reduction in RMSE is also consistently larger than the reduction in MAE, indicating that the Graph-Temporal model is particularly effective in reducing larger forecasting errors.

The same trend is illustrated in Figure 5, where the performance gap between the Temporal model and the Graph-Temporal model becomes progressively larger toward the 60-minute horizon.

5.2 Probabilistic Forecasting and Calibration

The probabilistic forecasting performance of the Temporal GRU and Graph-Temporal models is reported in Table 5. The Pinball Loss is evaluated separately for the three predicted quantiles, while empirical coverage and interval width are used to assess the reliability and sharpness of the nominal 80% prediction intervals.

Table 5: Probabilistic forecasting performance on the test set. Coverage refers to the nominal 80% prediction interval defined by $q_{0.1}$ and $q_{0.9}$.

Model	Horizon	Pinball $q_{0.1}$	Pinball $q_{0.5}$	Pinball $q_{0.9}$	Coverage (%)	Interval Width
Temporal GRU	15 min	0.952	1.540	0.643	81.30	9.339
	30 min	1.225	1.867	0.743	80.16	10.931
	60 min	1.668	2.367	0.832	79.81	13.309
Graph-Temporal	15 min	0.845	1.451	0.597	78.44	8.929
	30 min	1.028	1.721	0.677	78.22	10.485
	60 min	1.320	2.122	0.775	78.13	12.470

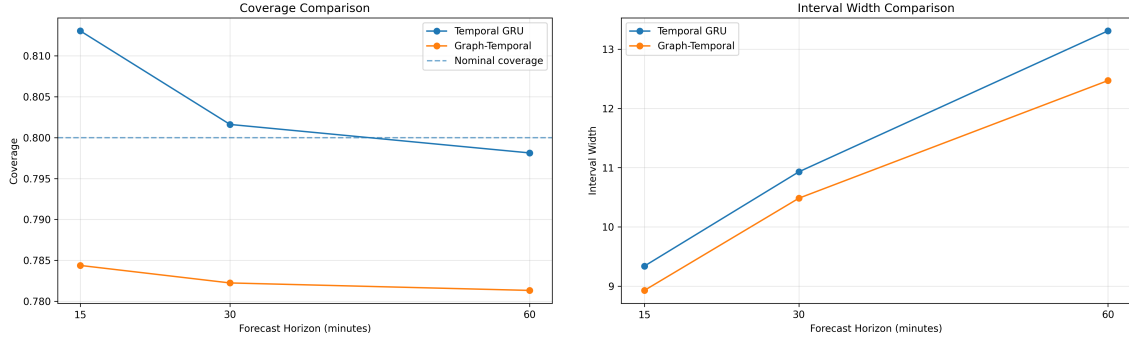


Figure 6: Comparison of the probabilistic forecasts produced by the Temporal GRU and Graph-Temporal models in terms of empirical coverage (left) and prediction interval width (right) across the three forecasting horizons. The dashed line in the coverage plot represents the nominal 80% coverage level.

Across all forecasting horizons, the Graph-Temporal model achieves lower Pinball Loss for each of the three quantiles. At 15 minutes, the losses for $q_{0.1}$, $q_{0.5}$, and $q_{0.9}$ decrease from 0.952, 1.540, and 0.643 for the Temporal GRU to 0.845, 1.451, and 0.597 for the Graph-Temporal model. The same behavior is observed at 30 and 60 minutes, indicating that the benefit of incorporating spatial information is not limited to the median forecast, but also extends to the estimation of the lower and upper quantiles.

The two models, however, exhibit a different trade-off between calibration and sharpness. The Temporal model achieves empirical coverage close to the nominal 80% level, with values of 81.30%, 80.16%, and 79.81% at 15, 30, and 60 minutes, respectively. The Graph-Temporal model produces narrower prediction intervals at every horizon, reducing the average interval width from 9.339 to 8.929 at 15 minutes, from 10.931 to 10.485 at 30 minutes, and from 13.309 to 12.470 at 60 minutes. This increased sharpness is accompanied by slightly lower coverage, which remains between 78.13% and 78.44% across the three horizons.

These results show that the Graph-Temporal model improves the quality of all three estimated quantiles and produces more concentrated predictive intervals, but the resulting intervals exhibit mild undercoverage relative to the nominal 80% level. In contrast, the Temporal model remains closer to the nominal coverage overall, although it exhibits mild overcoverage at the 15-minute horizon, reaching 81.30%. The reliability-sharpness trade-off is further illustrated in Figure 6. Moreover, interval width increases with the forecasting horizon for both models, reflecting the greater uncertainty of longer-term predictions. The calibration behavior of the selected Graph-Temporal model is further investigated under distribution shift in Section 7.

Quantile crossing is negligible for both models, with rates below 0.05% at every forecasting horizon, indicating that the predicted quantiles almost always preserve their expected ordering.

A more detailed view of probabilistic reliability is provided by the calibration curves in Figure 7. The plots compare the nominal quantile levels with their empirical frequencies and complement the interval-level coverage results reported in Table 5.

6 Analysis of the Graph Component

6.1 When Does the Graph Help?

The aggregate results presented in Section 5 show that incorporating graph information improves forecasting accuracy at all three horizons. To determine whether this improvement is widespread

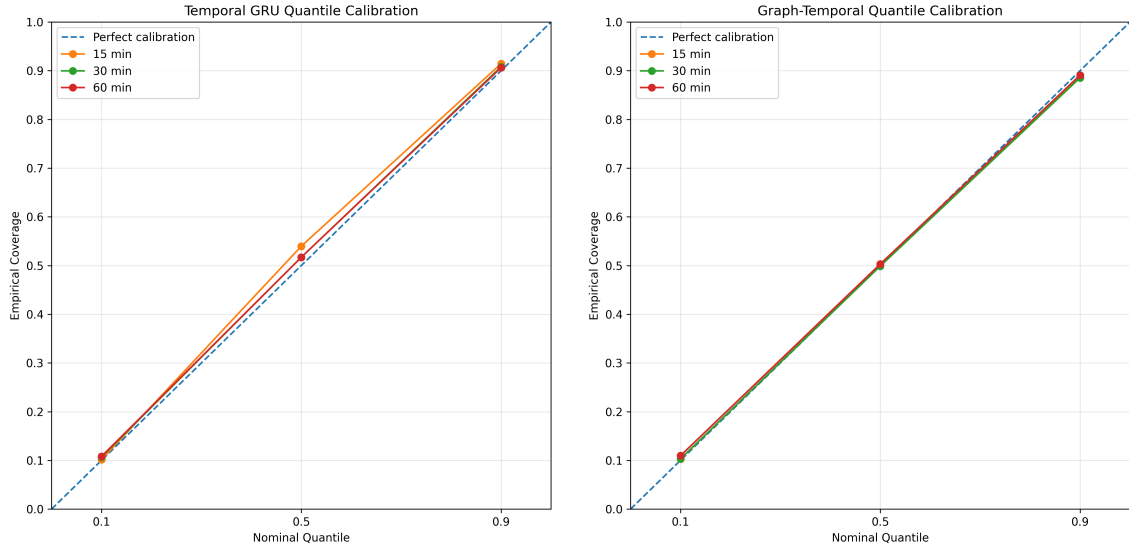


Figure 7: Quantile calibration of the Temporal (left) and Graph-Temporal (right) models at the three forecasting horizons. The dashed diagonal represents perfect calibration.

across the road network or concentrated on a limited subset of sensors, the two neural models were additionally compared at sensor level.

For each sensor n and forecasting horizon h , the effect of graph message passing is quantified as the difference between the MAE of the Temporal and that of the Graph-Temporal model,

$$\Delta\text{MAE}_{n,h} = \text{MAE}_{n,h}^{\text{Temporal}} - \text{MAE}_{n,h}^{\text{Graph}}. \quad (23)$$

A positive value of $\Delta\text{MAE}_{n,h}$ therefore indicates that incorporating graph information reduces the forecasting error for sensor n , whereas a negative value indicates that the Temporal model performs better for that sensor.

The sensor-level comparison shows that the aggregate improvement is not driven by only a small subset of locations. The Graph-Temporal model reduces MAE for 199 out of 207 sensors at the 15-minute horizon and for 200 out of 207 sensors at 30 minutes. At 60 minutes, graph message passing remains beneficial for 193 out of 207 sensors. Therefore, the advantage observed in the aggregate metrics is distributed across the large majority of the sensor network, although a small number of locations consistently or occasionally benefit more from purely temporal modeling.

To complement the sensor-level aggregate analysis, representative forecasting examples are examined in Figure 8. The examples compare the predictions of the Temporal and Graph-Temporal models for sensor-time pairs in which the inclusion of spatial information produces a clear reduction in forecasting error.

Figure 9 instead reports representative cases in which graph message passing increases the forecasting error. These examples are particularly useful because they show that neighboring information is not always informative for the future evolution of a given sensor. The presence of such cases motivates a more detailed investigation of whether the benefit of graph message passing can be related to the local graph structure.

Overall, these examples confirm that the effect of graph message passing is sensor and time dependent. Although spatial information improves forecasting accuracy for the large majority of sensors, it can occasionally introduce information that does not improve the local prediction. This observation is consistent with the small subset of sensors for which the Temporal GRU achieves lower MAE.

6.2 Effect of Diffusion Depth

The main Graph-Temporal model uses a diffusion depth of $K = 2$, allowing each diffusion-style message-passing layer to combine information from the current node and from nodes reached through one and two-step propagation. To investigate whether a larger spatial receptive field provides additional benefits, a variant with $K = 3$ was trained while keeping the remaining model configuration unchanged.

Table 6 compares the test performance of the two configurations. Increasing the diffusion depth produces a small but consistent reduction in MAE. However, the same behavior is not observed

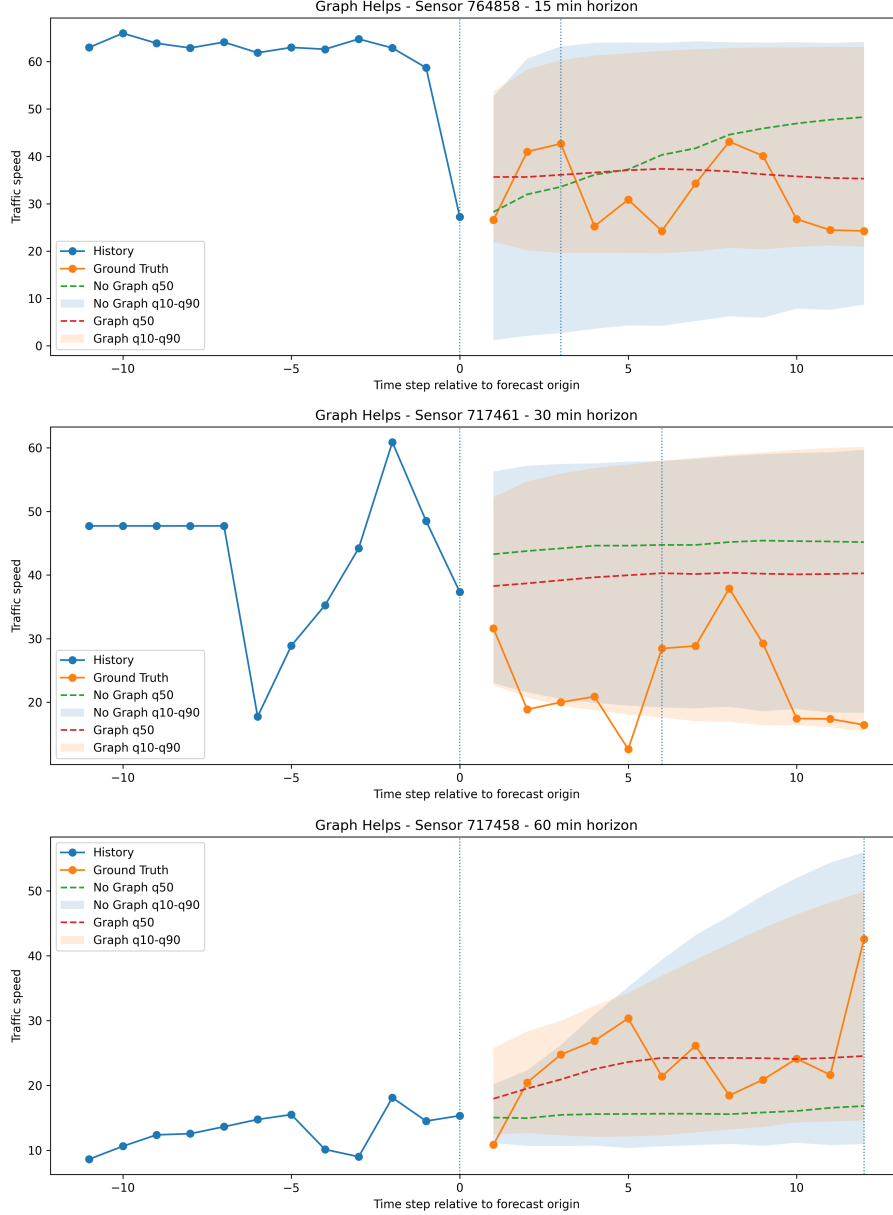


Figure 8: Representative forecasting examples in which the Graph-Temporal model improves over the Temporal GRU at the 15-minute (top), 30-minute (center), and 60-minute (bottom) forecasting horizons.

for RMSE since $K = 3$ model obtains slightly higher RMSE at all three horizons. Therefore, the additional diffusion step slightly improves the average absolute error but does not improve the model’s sensitivity to larger forecasting errors.

The comparison is also illustrated in Figure 10. The effect of the additional diffusion step is particularly visible in the probabilistic forecasts. The $K = 3$ model produces narrower prediction intervals at every horizon. At the shortest horizon, this increased sharpness is accompanied by a small improvement in coverage. At the longer forecasting horizons, however, the sharper intervals obtained with $K = 3$ come at the cost of increased undercoverage. Quantile crossing remains negligible for both configurations.

The Pinball Loss values show a similarly limited effect of increasing the diffusion depth. The $K = 3$ configuration slightly improves the median and upper-quantile losses at all horizons, while the loss associated with the lower quantile slightly increases. Overall, increasing the diffusion depth does not produce a consistent improvement across all probabilistic metrics.

The two configurations also exhibit different computational behavior. The $K = 2$ model was trained for 106 epochs, whereas the $K = 3$ variant stopped after 85 epochs. Despite requiring fewer epochs, the additional diffusion step increases the computational cost of each epoch: the average training time rises from approximately 19.3 seconds per epoch for $K = 2$ to 22.0 seconds for $K = 3$, an increase of about 14%. The total training times are approximately 34.0 and 31.1

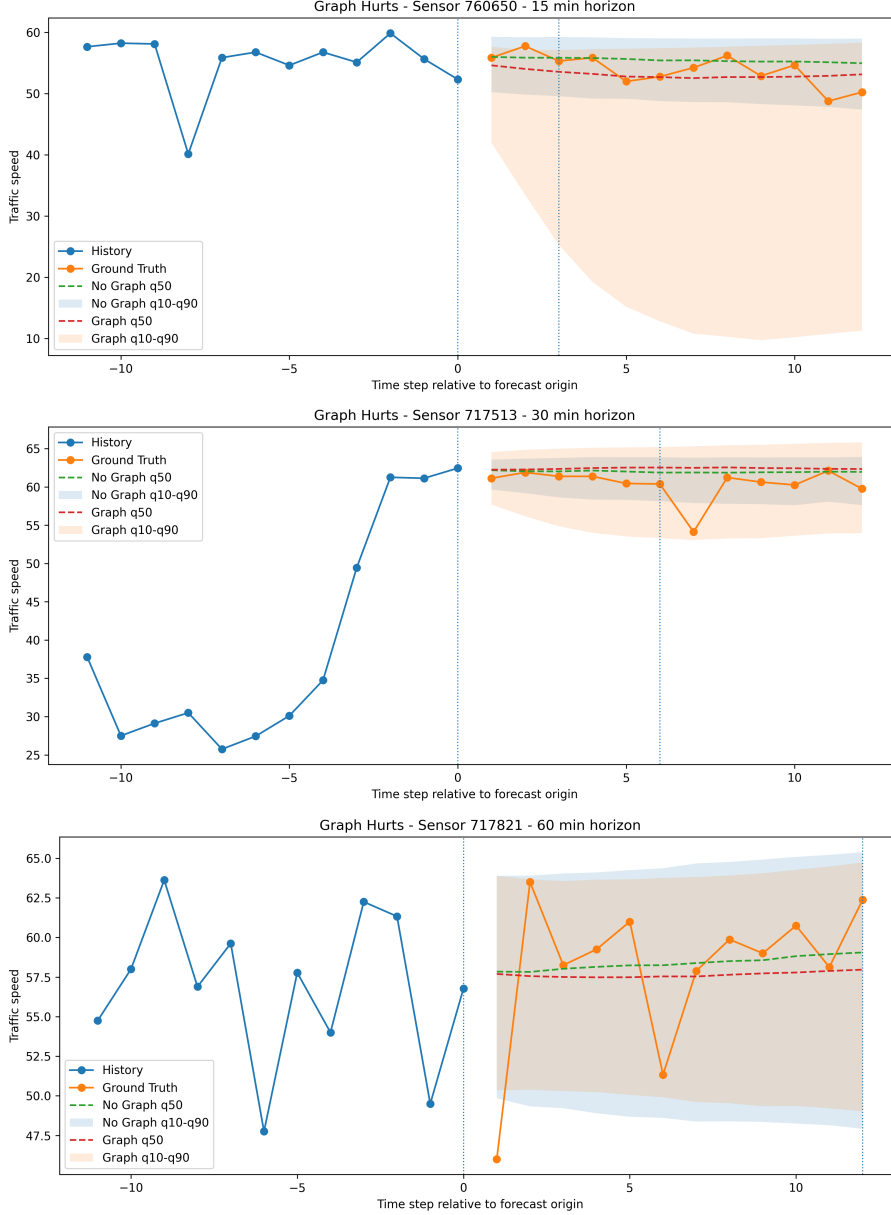


Figure 9: Representative forecasting examples in which the Graph-Temporal model performs worse than the Temporal model at the 15-minute (top), 30-minute (center), and 60-minute (bottom) forecasting horizons.

minutes, respectively, with the lower total training time of $K = 3$ resulting from earlier stopping rather than lower per-epoch computational complexity.

Overall, increasing the diffusion depth from $K = 2$ to $K = 3$ provides only modest improvements in MAE and median Pinball Loss, while slightly increasing RMSE and reducing coverage at the longer horizons. It also increases the computational cost of each training epoch. For these reasons, $K = 2$ was retained for the main Graph-Temporal model, providing a more balanced trade-off between forecasting accuracy, probabilistic reliability, and computational complexity.

6.3 Graph Ablation: Weighted vs. Binary Adjacency

The previous experiments use the original weighted adjacency matrix provided with METR-LA. To investigate whether the edge weights contribute to the forecasting performance or whether the graph connectivity alone is sufficient, an ablation experiment was performed by replacing every non-zero edge weight with one. The resulting binary adjacency matrix preserves the original graph topology but removes information about the relative strength of the connections. As for the weighted graph, the binary adjacency matrix was row-normalized before being used for diffusion message passing.

Table 6: Test performance of the Graph-Temporal model with diffusion depths $K = 2$ and $K = 3$.

K	Horizon	MAE	RMSE	Pinball _{0.1}	Pinball _{0.5}	Pinball _{0.9}	Coverage (%)	Interval Width	Crossing (%)
2	15 min	2.901	5.511	0.845	1.451	0.597	78.44	8.929	0.031
2	30 min	3.442	6.758	1.028	1.721	0.677	78.22	10.485	0.029
2	60 min	4.244	8.378	1.320	2.122	0.775	78.13	12.470	0.048
3	15 min	2.888	5.533	0.846	1.444	0.593	78.84	8.723	0.028
3	30 min	3.424	6.799	1.030	1.712	0.673	77.87	9.948	0.034
3	60 min	4.203	8.384	1.322	2.102	0.770	77.50	11.781	0.055

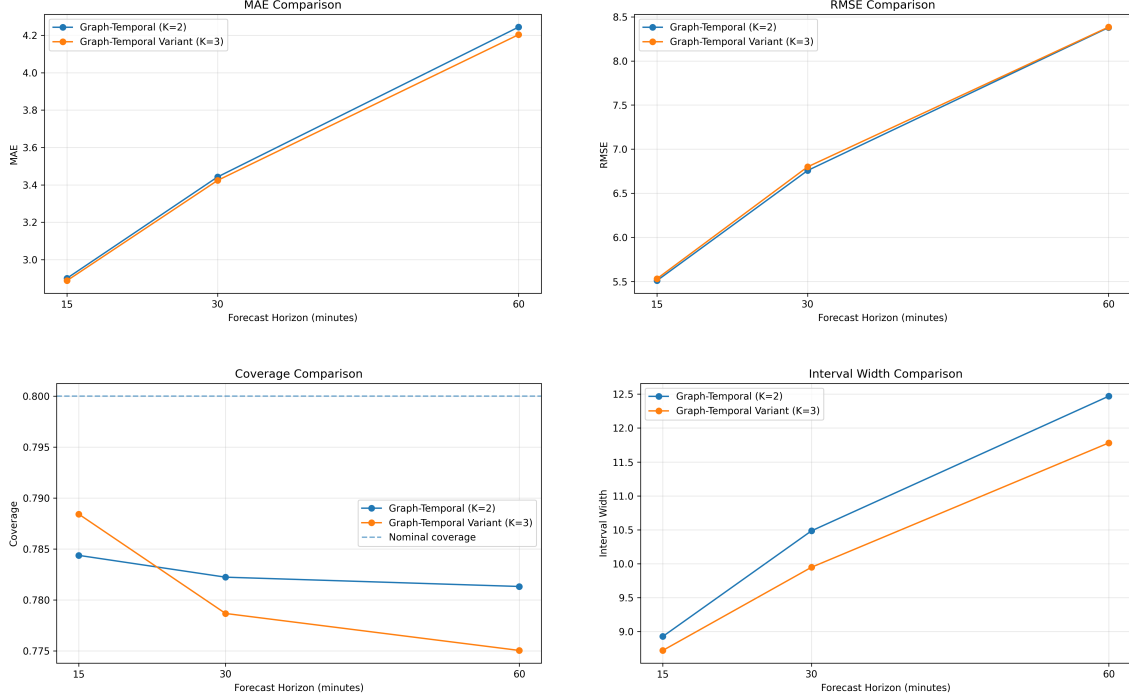
**Figure 10:** Comparison of the Graph-Temporal model with diffusion depths $K = 2$ and $K = 3$. The top row reports point forecasting performance in terms of MAE (left) and RMSE (right), while the bottom row reports empirical coverage (left) and prediction interval width (right) across the three forecasting horizons.

Table 7 compares the two graph representations on the test set. The weighted graph consistently achieves slightly lower point forecasting errors and the difference becomes larger as the forecasting horizon increases.

The comparison is further illustrated in Figure 11. The probabilistic results reveal a more nuanced difference. At 15 and 30 minutes, the binary graph produces slightly narrower intervals, while its empirical coverage is slightly closer to the nominal 80% level. At 60 minutes, however, the binary representation produces a slightly wider interval, while coverage remains almost unchanged. Therefore, binarizing the graph does not provide a consistent advantage in terms of interval sharpness or reliability.

The probabilistic metrics provide further evidence in favor of preserving the original edge weights. The weighted graph achieves lower Pinball Loss for all three predicted quantiles and at every forecasting horizon. The difference also becomes more pronounced at longer horizons, consistently with the behavior observed for MAE and RMSE. This indicates that the edge weights contribute not only to point forecasting accuracy, but also to the estimation of the predictive quantiles.

Overall, these results indicate that the benefit of graph message passing is not determined solely by the presence of connections between sensors. The relative edge weights provide useful information for spatial aggregation, with their contribution becoming more evident as the forecasting horizon increases. Since the binary graph does not provide a consistent improvement in probabilistic reliability and results in higher point errors, the original weighted adjacency matrix was retained for the main Graph-Temporal model.

Table 7: Test performance of the Graph-Temporal model using the weighted and binary adjacency matrices.

Graph	Horizon	MAE	RMSE	Pinball _{0.1}	Pinball _{0.5}	Pinball _{0.9}	Coverage (%)	Interval Width
Weighted	15 min	2.901	5.511	0.845	1.451	0.597	78.44	8.929
	30 min	3.442	6.758	1.028	1.721	0.677	78.22	10.485
	60 min	4.244	8.378	1.320	2.122	0.775	78.13	12.470
Binary	15 min	2.948	5.682	0.873	1.474	0.610	79.54	8.878
	30 min	3.543	7.033	1.084	1.772	0.699	78.80	10.424
	60 min	4.443	8.754	1.417	2.222	0.807	78.11	12.593

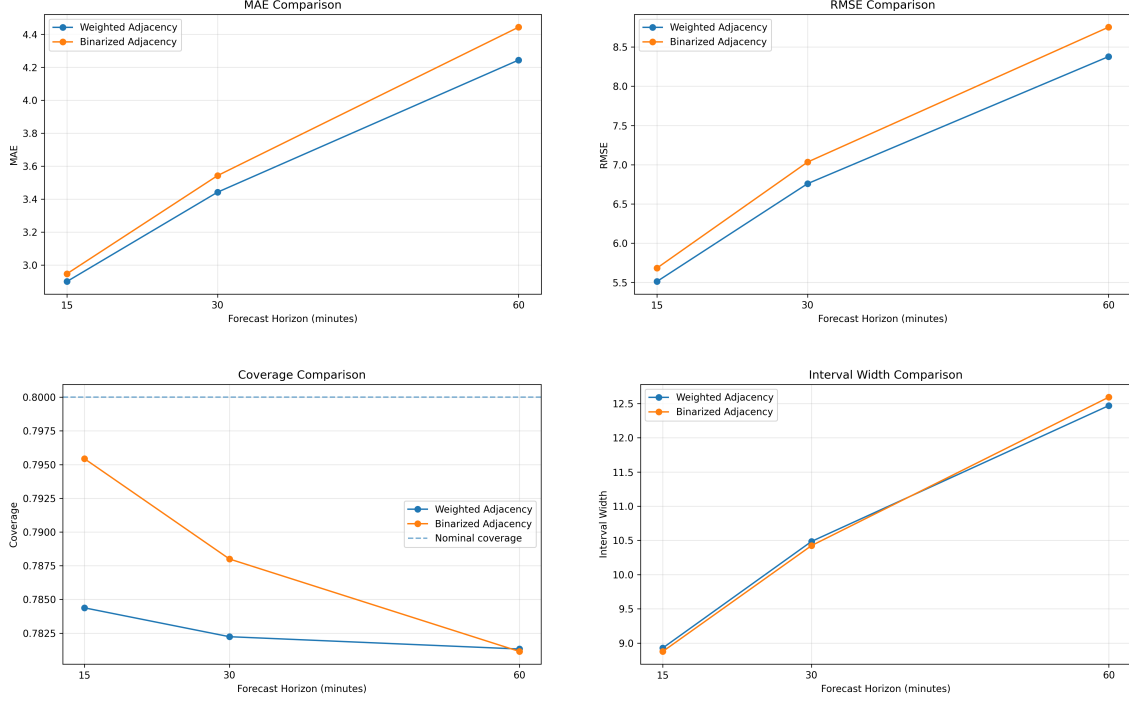


Figure 11: Comparison of the Graph-Temporal model using the weighted and binary adjacency matrices. The top row reports point forecasting performance in terms of MAE (left) and RMSE (right), while the bottom row reports empirical coverage (left) and prediction interval width (right) across the three forecasting horizons.

7 Robustness under Distribution Shift

The previous experiments evaluate forecasting performance over the complete test distribution. However, average performance may hide substantial degradation under traffic conditions that are less frequently represented during training. To investigate this aspect, the Graph-Temporal model is evaluated under a predefined high-congestion stress condition. The stress condition is defined exclusively from the training data, preventing information from the validation or test sets from influencing the definition of the shifted subset.

7.1 High-Congestion Definition

High-congestion conditions are identified using sensor-specific speed thresholds. For each sensor n , a congestion threshold τ_n is defined as the 10th percentile of its valid speed observations in the training set,

$$\tau_n = Q_{0.10}(\{y_{t,n}^{\text{train}} : m_{t,n}^{\text{train}} = 1\}), \quad (24)$$

where $m_{t,n}^{\text{train}}$ denotes the availability mask. A sensor is therefore considered congested at time t when its valid observed speed is lower than or equal to its sensor-specific threshold. Using a separate threshold for each sensor accounts for the different typical speed ranges observed across locations in the road network.

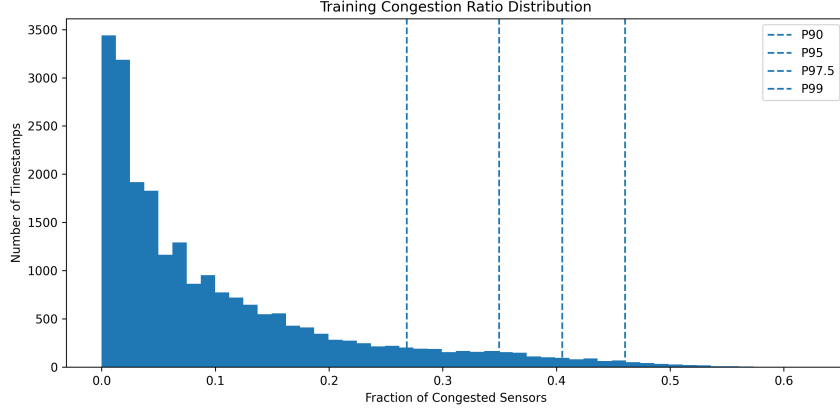


Figure 12: Distribution of the network-level congestion ratio C_t in the training set.

The network-level congestion ratio at timestamp t is then defined as the fraction of valid sensors that are classified as congested,

$$C_t = \frac{\sum_{n=1}^N m_{t,n} \mathbb{I}(y_{t,n} \leq \tau_n)}{\sum_{n=1}^N m_{t,n}}, \quad (25)$$

where $\mathbb{I}(\cdot)$ is the indicator function. This definition measures the spatial extent of congestion at each timestamp rather than relying on the speed of a single sensor.

Finally, the threshold used to identify a high-congestion event is defined as the 95th percentile of the congestion ratios observed in the training set,

$$\tau_{\text{peak}} = Q_{0.95}(\{C_t^{\text{train}}\}). \quad (26)$$

For the METR-LA training partition, this procedure yields $\tau_{\text{peak}} = 0.3495$. Consequently, a timestamp is classified as a high-congestion condition when at least approximately 35% of the valid sensors simultaneously exhibit speeds below their respective training-based 10th-percentile thresholds.

Figure 12 shows the training distribution used to define the stress condition. By selecting its 95th percentile, the stress test focuses on relatively uncommon periods in which congestion simultaneously affects a substantial fraction of the network. Importantly, both the sensor-specific thresholds τ_n and the network-level threshold τ_{peak} are estimated exclusively from the training partition and subsequently kept fixed when classifying validation and test observations.

7.2 Stress-Test Protocol

The high-congestion definition introduced in Section 7.1 is applied to the forecasting targets of each test window. Since the model is evaluated at three different forecasting horizons, congestion is assessed independently at the target timestamp corresponding to each horizon.

More specifically, for a test sample whose input sequence ends at time t , the congestion condition for forecasting horizon h is determined from the network-level congestion ratio at the corresponding future timestamp $t + h$. The sample is classified as high congestion for horizon h when

$$C_{t+h} \geq \tau_{\text{peak}}, \quad (27)$$

where $\tau_{\text{peak}} = 0.3495$ is the threshold estimated exclusively from the training partition. Otherwise, the sample is assigned to the normal subset for that horizon.

This horizon-specific classification is necessary because traffic conditions may change within the 60-minute forecasting window. Consequently, the same input window can be classified as normal at one forecasting horizon and as high congestion at another. The stress-test subsets for 15, 30, and 60 minutes should therefore be interpreted independently rather than as a single fixed set of test windows.

Applying this procedure to the 6224 test windows yields 484 high-congestion samples and 5740 normal samples at each evaluated horizon. Although the subset sizes are identical, their composition differs across horizons because the congestion condition is evaluated independently at the target timestamp associated with each forecast horizon. Consequently, the same input window may belong to different traffic-condition subsets at 15, 30, and 60 minutes.

Table 8: Performance of the Graph-Temporal model under normal and high-congestion conditions on the test set.

Condition	Horizon	MAE	RMSE	Pinball _{0.1}	Pinball _{0.5}	Pinball _{0.9}	Coverage (%)	Interval Width
Normal	15 min	2.770	5.289	0.819	1.385	0.559	78.4	8.499
	30 min	3.257	6.453	0.990	1.629	0.626	78.3	9.911
	60 min	3.974	7.957	1.266	1.987	0.705	78.3	11.730
High congestion	15 min	4.463	7.669	1.159	2.232	1.044	78.7	14.035
	30 min	5.636	9.660	1.467	2.818	1.281	77.9	17.283
	60 min	7.414	12.296	1.957	3.707	1.596	75.8	21.169

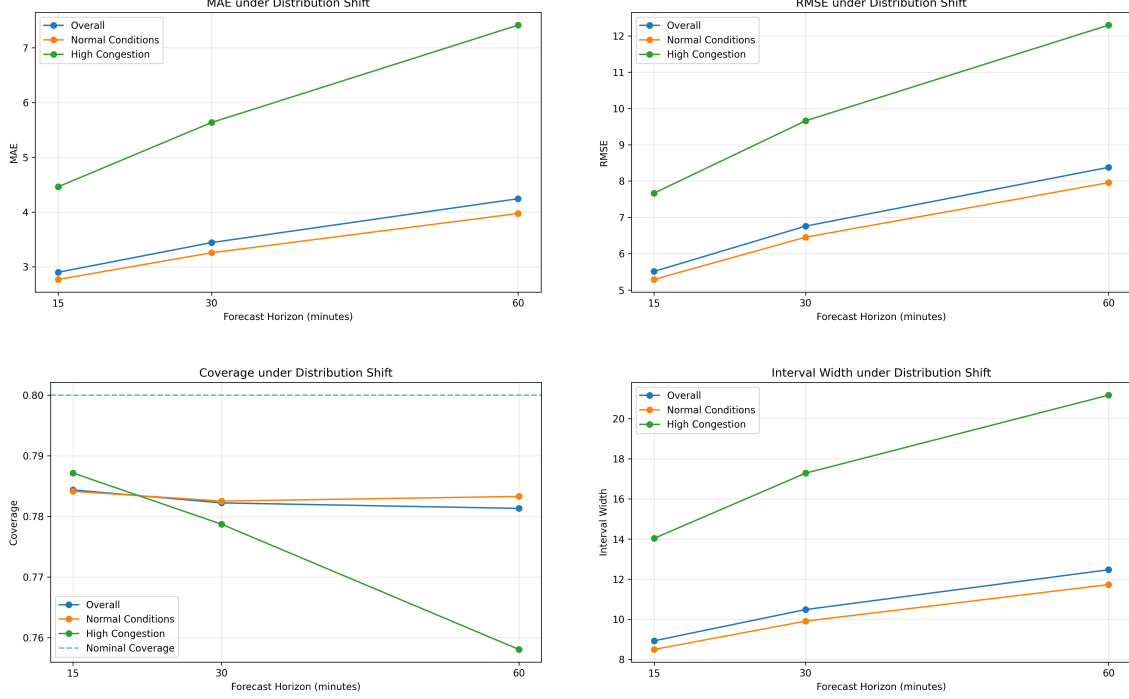


Figure 13: Performance of the Graph-Temporal model under normal and high-congestion conditions. The top row compares MAE (left) and RMSE (right), while the bottom row reports empirical coverage (left) and prediction interval width (right) across the three forecasting horizons.

The stress test evaluates the same trained Graph-Temporal model used in the previous experiments, without retraining or adapting it to high-congestion conditions. Forecasting metrics are computed separately over the normal and high-congestion subsets using the target availability mask. This allows the degradation associated with the shifted traffic condition to be measured while keeping the model, preprocessing pipeline, and evaluation procedure unchanged.

7.3 Quantitative Results

Table 8 reports the performance of the Graph-Temporal model separately under normal and high-congestion conditions. A substantial degradation is observed under high congestion at all three forecasting horizons, and the difference becomes progressively larger as the prediction horizon increases.

It is worth noting that the performance observed under normal conditions is close to the overall test performance reported in Section 5. This is expected from the definition of the stress condition. High congestion is intentionally defined as a relatively rare event through the 95th percentile of the training congestion-ratio distribution and accounts for only 484 out of 6224 test windows, approximately 7.8%, at each forecasting horizon. Consequently, the overall test metrics are dominated by the much larger normal subset and can therefore conceal the substantially poorer performance observed during high-congestion periods.

The degradation in point forecasting accuracy is clearly visible in Figure 13. At 15 minutes, MAE increases from 2.770 under normal conditions to 4.463 under high congestion, corresponding to an increase of approximately 61%. The difference becomes larger at longer horizons: MAE increases from 3.257 to 5.636 at 30 minutes and from 3.974 to 7.414 at 60 minutes, corresponding to increases

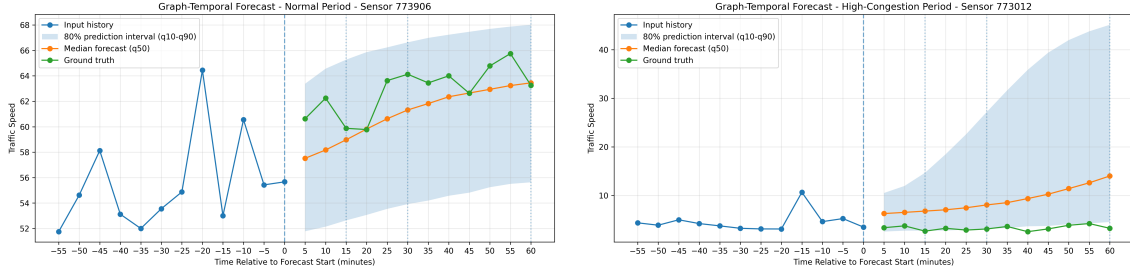


Figure 14: Representative probabilistic forecasts produced by the Graph-Temporal model under normal (left) and high-congestion (right) conditions.

of approximately 73% and 87%, respectively. RMSE exhibits the same behavior, increasing from 5.289 to 7.669, from 6.453 to 9.660, and from 7.957 to 12.296. These results show that forecasting becomes substantially more difficult during network-wide congestion and that the performance degradation becomes more pronounced as the forecasting horizon increases.

The prediction intervals respond to the increased forecasting difficulty by becoming substantially wider under high congestion. Their average width increases from 8.499 to 14.035 at 15 minutes, from 9.911 to 17.283 at 30 minutes, and from 11.730 to 21.169 at 60 minutes. The model therefore captures part of the increased uncertainty associated with congested traffic conditions rather than maintaining intervals of approximately constant width.

However, the increase in interval width is not sufficient to preserve calibration at the longest forecasting horizon. At 15 minutes, empirical coverage remains approximately stable, changing from 78.4% under normal conditions to 78.7% under high congestion. At 30 minutes, it decreases slightly from 78.3% to 77.9%. The largest degradation occurs at 60 minutes, where coverage decreases from approximately 78.3% to 75.8%, despite the prediction intervals being substantially wider.

The pronounced increase in interval width also represents a loss of sharpness. Although wider intervals appropriately express greater predictive uncertainty, excessively wide intervals provide a less precise and therefore less informative characterization of future traffic conditions. Under high congestion, the model therefore exhibits a double degradation at longer horizons: the prediction intervals become less sharp while still failing to achieve the nominal 80% coverage. This indicates that the additional uncertainty expressed by the model is not sufficient to maintain reliable probabilistic forecasts under the shifted traffic regime.

The degradation under high congestion is also evident in the probabilistic forecasting errors. Pinball Loss increases for all three predicted quantiles and at every forecasting horizon, with the differences becoming larger at longer horizons. This shows that high congestion affects not only the median forecast, but the estimated predictive distribution more generally.

Overall, the stress test reveals two distinct effects of high-congestion conditions. First, point forecasting errors increase substantially, showing that these traffic regimes are more difficult to predict than normal conditions. Second, although the model reacts to this increased difficulty by producing wider prediction intervals, the uncertainty estimates do not fully compensate for the degradation at longer horizons. In particular, the combination of substantially wider intervals and lower coverage at 60 minutes indicates residual overconfidence under distribution shift. This motivates the mitigation strategies investigated in Section 8, which address the observed degradation through both post-hoc calibration and congestion-aware training.

7.4 Qualitative Analysis

The aggregate behavior observed in the stress test is further examined through representative forecasting examples. Figure 14 compares predictions produced by the Graph-Temporal model under normal and high-congestion conditions. The examples are intended to illustrate how the changes in forecasting accuracy and predictive uncertainty identified in the quantitative analysis appear at the level of individual forecasting windows.

Figure 14 provides a qualitative view of how the forecasting behavior changes between normal and high-congestion conditions. In the normal example (left), traffic speeds remain relatively high and the model successfully captures the increasing trend observed over the forecasting window. Although the median forecast does not reproduce every short-term fluctuation of the ground truth, it follows its overall evolution, while the prediction interval remains moderately wide and contains the observed values throughout most of the horizon.

A markedly different behavior emerges in the high-congestion example (right). The very low observed speeds reflect a strongly congested traffic regime. The median forecast remains relatively close to the ground truth at the beginning of the forecasting window, but progressively overestimates future traffic speed as the horizon increases. At the same time, the prediction interval is substantially wider than in the normal example and expands further toward longer horizons. This widening indicates that the model recognizes the increasing uncertainty associated with forecasting the evolution of the congested state. Nevertheless, the ground-truth values remain below the lower prediction bound for most of the forecasting window. Therefore, even though the model expresses considerably greater uncertainty, the predictive interval is not sufficiently well positioned to cover the actual traffic evolution.

These examples illustrate the two effects identified quantitatively in Section 7.3. High congestion increases the difficulty of the point forecasting task and simultaneously leads the model to produce wider uncertainty intervals. Although wider intervals can reflect greater uncertainty, excessively wide intervals are also less informative, as they provide a less precise characterization of the range of plausible future traffic conditions. Moreover, increasing interval width alone is not sufficient to guarantee reliable uncertainty estimates when the predictive distribution is systematically displaced from the observations. This behavior provides a qualitative explanation for the reduced empirical coverage observed under high congestion, particularly at longer forecasting horizons, and highlights the need to consider both coverage and interval sharpness when assessing probabilistic forecasts.

8 Mitigation Strategies

The distribution-shift analysis showed that the Graph-Temporal model increases the width of its prediction intervals during high-congestion periods, but this adaptation is not sufficient to preserve the desired coverage, particularly at the 60-minute forecasting horizon. This section investigates two complementary mitigation strategies. First, a post-hoc calibration procedure is applied to improve the reliability of the predictive intervals without retraining the forecasting model. Second, a congestion-aware training objective is evaluated to explicitly increase the importance of high-congestion targets during model optimization.

8.1 Conformalized Quantile Regression

Conformalized Quantile Regression (CQR) [2] was applied as a post-hoc calibration procedure to the prediction intervals produced by the trained Graph-Temporal model. The method uses the validation set as a calibration set and adjusts the lower and upper predicted quantiles according to their observed nonconformity with respect to the ground truth. The median prediction is left unchanged, so the procedure affects uncertainty estimation without modifying the point forecast. For each valid calibration target y_i , with predicted lower and upper quantiles $\hat{q}_{0.1,i}$ and $\hat{q}_{0.9,i}$, the nonconformity score is defined as

$$s_i = \max(\hat{q}_{0.1,i} - y_i, y_i - \hat{q}_{0.9,i}). \quad (28)$$

The score measures how far an observation lies outside the predicted interval. Observations already contained within the interval produce non-positive scores, whereas positive scores quantify the amount by which the interval fails to cover the target.

The calibration scores are then used to determine how much the original prediction intervals should be enlarged. Since the target coverage is 80%, the desired miscoverage level is $\alpha = 0.2$. Let n_{cal} denote the number of valid calibration scores. The correction $\hat{s}$ is selected as the empirical quantile of the calibration scores at level

$$q = \frac{\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil}{n_{\text{cal}}}, \quad (29)$$

and therefore

$$\hat{s} = Q_q(\{s_i\}_{i=1}^{n_{\text{cal}}}). \quad (30)$$

Intuitively, $\hat{s}$ represents the conformal correction applied to the original interval bounds to achieve the target coverage level on the calibration data.

The calibrated prediction interval for a test observation is obtained by expanding both bounds by this correction,

Table 9: Empirical coverage before and after CQR calibration under overall, normal, and high-congestion test conditions.

Condition	Horizon	Before CQR (%)	After CQR (%)
Overall	15 min	78.44	80.24
	30 min	78.22	80.14
	60 min	78.13	79.96
Normal	15 min	78.41	80.27
	30 min	78.25	80.23
	60 min	78.33	80.21
High congestion	15 min	78.72	79.89
	30 min	77.87	79.07
	60 min	75.81	77.04

$$[\hat{q}_{0.1,i} - \hat{s}, \hat{q}_{0.9,i} + \hat{s}]. \quad (31)$$

The procedure is performed separately for each forecasting horizon, allowing the calibration correction to reflect the different uncertainty levels at 15, 30, and 60 minutes. All calibration quantities are estimated exclusively from the validation partition and are subsequently kept fixed during test evaluation. Therefore, no test targets are used to determine the conformal correction.

CQR therefore provides a lightweight mitigation strategy: it does not alter the Graph-Temporal architecture, its learned parameters, or its median forecast, but only recalibrates the predicted uncertainty intervals. This makes it particularly suitable for the failure mode identified in Section 7, where the model already reacts to high-congestion conditions by increasing interval width, but still exhibits residual undercoverage.

8.1.1 Calibration Results

The effect of CQR is evaluated by comparing the empirical coverage and prediction interval width before and after calibration. Table 9 reports the results for the complete test set and separately for normal and high-congestion conditions.

As shown in Table 9, CQR effectively corrects the mild undercoverage observed on the complete test distribution. Empirical coverage increases from 78.44%, 78.22%, and 78.13% to 80.24%, 80.14%, and 79.96% at the 15, 30, and 60-minute horizons, respectively. The calibrated intervals are therefore very close to the nominal 80% coverage level across all three forecasting horizons.

A similar behavior is observed under normal traffic conditions, where coverage after calibration reaches 80.27%, 80.23%, and 80.21%. This is consistent with the fact that normal conditions represent the dominant operating regime in both the calibration and test distributions. CQR is therefore effective at correcting the systematic in-distribution miscalibration of the original prediction intervals.

The improvement is more limited under high congestion. Coverage increases from 78.72% to 79.89% at 15 minutes and from 77.87% to 79.07% at 30 minutes, bringing both substantially closer to the nominal level. At 60 minutes, however, coverage increases only from 75.81% to 77.04% and remains clearly below the desired 80%. Thus, the same calibration procedure that provides nearly nominal coverage overall does not fully correct the undercoverage observed in the most challenging shifted condition.

Figure 15 shows that the improvement in coverage is obtained by further expanding the original prediction intervals. This introduces an important trade-off, since the intervals were already substantially wider under high-congestion conditions. Although CQR improves empirical coverage, the additional widening further reduces interval sharpness and can therefore make the probabilistic forecasts less informative. Since CQR leaves the median forecast unchanged, it does not address the increase in point forecasting error observed under high congestion.

The remaining gap under high congestion, particularly at 60 minutes, highlights an important limitation of the global calibration strategy.

Overall, CQR successfully mitigates the general undercoverage of the Graph-Temporal model and substantially improves calibration under normal conditions. It also improves coverage during high congestion, but does not completely eliminate the reliability degradation observed at the longest forecasting horizon. This residual failure motivates considering mitigation strategies that explicitly account for congestion during model training.

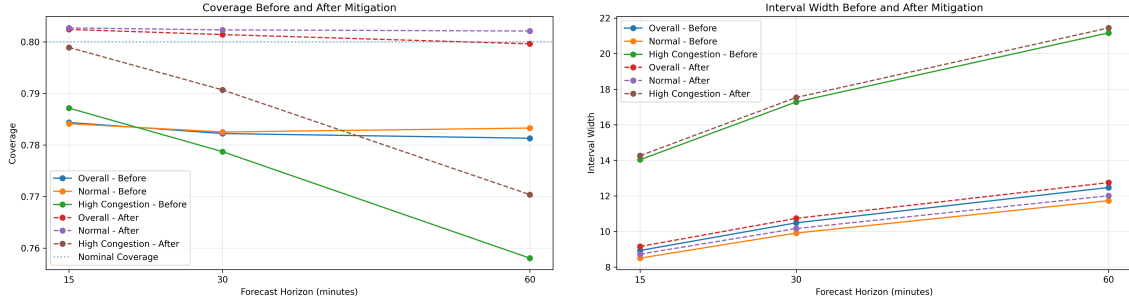


Figure 15: Effect of CQR calibration on empirical coverage (left) and prediction interval width (right) under overall, normal, and high-congestion conditions.

8.2 Congestion-Aware Training

Although CQR improves the calibration of the prediction intervals, the remaining undercoverage under high-congestion conditions suggests that a global post-hoc correction cannot completely address the distribution shift. As a second mitigation strategy, the training objective was therefore modified to explicitly increase the importance of high-congestion targets. Previous work on traffic speed forecasting has addressed the imbalance between free-flow and congested conditions through congestion-adaptive loss weighting, assigning greater importance to prediction errors occurring under congested traffic states [3].

Following this principle, a congestion-aware version of the masked Pinball Loss was implemented. The congestion condition was defined using the same training-derived criterion introduced in Section 7. For each target timestamp, the network congestion ratio C_{t+h} was computed from the fraction of valid sensors whose speed was below their sensor-specific congestion threshold. A target timestamp was classified as high congestion when

$$C_{t+h} \geq \tau_{\text{peak}}, \quad (32)$$

where τ_{peak} is the network-level congestion threshold estimated exclusively from the training data. A congestion-dependent weight was then assigned to each forecast step as

$$w_{b,h} = 1 + \lambda \mathbb{I}(C_{t+h} \geq \tau_{\text{peak}}), \quad (33)$$

where b denotes the sample, h the forecast step, and λ controls the additional importance assigned to high-congestion targets. In the implemented configuration, $\lambda = 1$ was used, so that normal targets received weight 1 and high-congestion targets received weight 2.

The resulting congestion-aware training objective is

$$\mathcal{L}_{\text{CA}} = \frac{\sum_{b,h,n} w_{b,h} m_{b,h,n} \sum_{q \in \mathcal{Q}} \rho_q(y_{b,h,n} - \hat{y}_{b,h,n}^{(q)})}{|\mathcal{Q}| \sum_{b,h,n} w_{b,h} m_{b,h,n}}, \quad (34)$$

where $m_{b,h,n}$ is the target-availability mask, $\mathcal{Q} = \{0.1, 0.5, 0.9\}$ is the set of predicted quantiles, and $\rho_q(\cdot)$ denotes the Pinball Loss for quantile q . The congestion weight is shared across sensors and quantiles for a given sample and forecast step, while missing target values remain excluded through the availability mask.

The weighting was applied independently to all 12 forecast steps used during training. The same congestion-aware objective was also used on the validation set for early stopping. The Graph-Temporal architecture was left unchanged, so the mitigation modifies only the optimization objective and does not add model parameters. With $\lambda = 1$, the best validation checkpoint was obtained at epoch 75.

Unlike CQR, congestion-aware training directly modifies the forecasting model by emphasizing the traffic regime in which the largest performance degradation was observed. However, the congestion information is required only during training and validation to construct the loss weights. At inference time, the trained model uses the same historical traffic inputs and graph structure as the original Graph-Temporal model and does not require knowledge of future congestion conditions. The effect of this training-time mitigation is evaluated in the following section.

8.2.1 Congestion-Aware Training Results

Table 10: Performance of the original and congestion-aware Graph-Temporal models under high-congestion conditions.

Metric	15 min		30 min		60 min	
	Original	CA	Original	CA	Original	CA
MAE	4.463	4.410	5.636	5.483	7.414	7.012
RMSE	7.669	7.655	9.660	9.567	12.296	11.883
Coverage (%)	78.72	79.39	77.87	78.34	75.81	76.98
Interval width	14.035	13.672	17.283	16.604	21.169	20.600

Table 10 directly compares the original Graph-Temporal model with the congestion-aware variant in the regime targeted by the reweighting strategy. Under high congestion, MAE decreases from 4.463 to 4.410 at 15 minutes, from 5.636 to 5.483 at 30 minutes, and from 7.414 to 7.012 at 60 minutes. The relative improvement becomes more pronounced as the forecasting horizon increases, reaching approximately 5.4% at 60 minutes. A similar trend is observed for RMSE, with the largest reduction occurring at the 60-minute horizon, from 12.296 to 11.883.

The probabilistic forecasts also improve in the targeted regime. Empirical coverage increases from 78.72% to 79.39% at 15 minutes, from 77.87% to 78.34% at 30 minutes, and from 75.81% to 76.98% at 60 minutes. The mitigation therefore reduces the undercoverage observed in the original Graph-Temporal model, although it does not completely restore the nominal 80% coverage at the longest forecasting horizon.

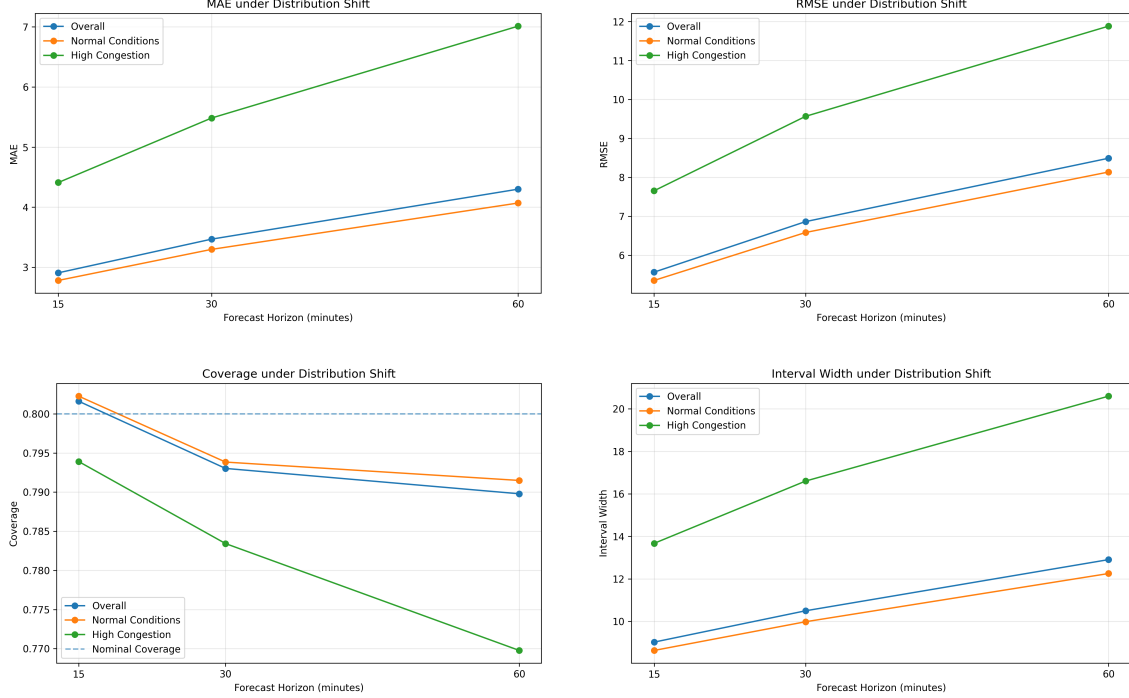
Importantly, the improvement in coverage is not obtained by simply widening the prediction intervals. Under high congestion, interval width decreases from 14.035 to 13.672 at 15 minutes, from 17.283 to 16.604 at 30 minutes, and from 21.169 to 20.600 at 60 minutes. The congestion-aware model therefore achieves higher coverage while producing sharper intervals. Together with the reductions in MAE and RMSE, this indicates that the reweighted training objective improves the forecasts in the shifted regime rather than merely increasing the amount of predicted uncertainty. Figure 16 provides a complementary view of the congestion-aware model by comparing its performance over the complete test set, normal traffic conditions, and high-congestion periods. The MAE and RMSE plots show that high congestion remains substantially more difficult than normal traffic and that the performance gap increases with the forecast horizon. In particular, point forecasting errors increase considerably at 60 minutes under high congestion, despite the improvements over the original model reported in Table 10.

The same distinction between traffic regimes is visible in the probabilistic predictions. As shown in the coverage plot of Figure 16, coverage remains close to the nominal 80% level at 15 minutes but decreases with the forecasting horizon, especially under high congestion. At the same time, the interval-width plot shows that prediction intervals become progressively wider for longer horizons and are substantially wider during congestion than under normal conditions. This indicates that the model appropriately expresses greater uncertainty during difficult traffic regimes, although the increase in interval width is still insufficient to maintain nominal coverage at the longest horizon. The improvements under congestion are accompanied by a moderate trade-off under normal traffic conditions. Normal-condition MAE increases from 2.770 to 2.780 at 15 minutes, from 3.257 to 3.297 at 30 minutes, and from 3.974 to 4.069 at 60 minutes relative to the original Graph-Temporal model. Emphasizing high-congestion targets during optimization therefore slightly reduces point-forecasting accuracy in the dominant normal regime. The effect is limited at short horizons but becomes more visible at 60 minutes.

Overall, the two implemented mitigation strategies address different aspects of the robustness problem. CQR provides a lightweight post-hoc correction that brings marginal coverage close to the nominal level without modifying the median forecasts. Congestion-aware training instead modifies the learned forecasting function itself and improves both point accuracy and probabilistic performance within the high-congestion regime, at the cost of a moderate degradation in normal-condition point accuracy. Neither strategy completely eliminates the long-horizon calibration gap under severe congestion, but the results indicate that global post-hoc calibration and regime-aware training provide complementary forms of mitigation.

Table 11: Computational cost of the Temporal and Graph-Temporal models.

Model	Trainable Parameters	Training Time (min)
Temporal GRU	40164	14.3
Graph-Temporal	64868	34.0

**Figure 16:** Performance of the congestion-aware Graph-Temporal model across the evaluated forecasting horizons. The top row reports point forecasting accuracy in terms of MAE (left) and RMSE (right), while the bottom row reports empirical coverage (left) and prediction interval width (right). Results are shown for the overall test set, normal traffic conditions, and high-congestion periods.

9 Computational Cost and Accuracy-Reliability Trade-off

The introduction of graph-based message passing improves forecasting accuracy, but also increases model complexity and computational cost. Table 11 compares the Temporal model and the selected Graph-Temporal model in terms of number of trainable parameters and training time.

As shown in Table 11, the Graph-Temporal model contains 64868 trainable parameters compared with 40164 for the Temporal model, corresponding to an increase of approximately 61.5%. Training time also increases from approximately 14.3 to 34.0 minutes, making the Graph-Temporal model about 2.4 times more expensive to train in the considered experimental setting. This additional cost results from the repeated multi-hop graph propagation operations and from the stacked graph-temporal blocks used to combine spatial and temporal information.

The additional computational cost is accompanied by improved forecasting accuracy. As shown in Section 5, the Graph-Temporal model consistently outperforms the Temporal model in terms of MAE, RMSE, and Pinball Loss, with the improvement in point forecasting becoming more pronounced at longer horizons. It also produces sharper prediction intervals, although with slightly lower empirical coverage. Therefore, the accuracy gains provided by graph message passing come at the cost of both increased computation and mild undercoverage.

Moreover, the stress-test analysis shows that this reliability gap becomes more important under high congestion. Although the Graph-Temporal model responds to difficult traffic conditions by producing wider intervals, coverage decreases to 75.8% at the 60-minute horizon. The model therefore does not fully translate its increased uncertainty into reliable coverage under distribution shift.

The mitigation experiments further illustrate different accuracy-reliability-cost trade-offs. CQR provides an inexpensive post-hoc mechanism for improving marginal calibration, while congestion-aware training requires retraining but directly improves forecasting performance in the high-

congestion regime. The latter reduces both point error and interval width while increasing coverage under congestion, although this comes with a small loss of accuracy under normal conditions. These results reinforce the need to evaluate robustness jointly in terms of accuracy, calibration, sharpness, and computational cost.

Overall, the Graph-Temporal architecture provides a clear accuracy advantage over the purely temporal baseline at the cost of greater computational complexity. Its probabilistic forecasts are generally sharper and more accurate in terms of quantile loss, but require additional calibration to approach the desired coverage. The results therefore highlight that model selection should consider not only point forecasting accuracy, but also computational cost and the reliability of predictive uncertainty, particularly under distribution shift.

10 Discussion

10.1 Main Findings

The experiments provide evidence that both temporal and spatial information are relevant for traffic forecasting on METR-LA. The Temporal model already provides a substantial improvement over persistence, showing that recent sensor histories contain strong predictive information. Incorporating graph message passing further improves forecasting accuracy, with the benefit becoming more evident at longer forecasting horizons. Moreover, the sensor-level analysis shows that this improvement is widespread across the network, although a small number of sensors do not benefit from spatial information.

The graph ablations further show that the way spatial information is represented matters. Increasing the diffusion depth from $K = 2$ to $K = 3$ provides only limited additional benefits and introduces a less favorable balance between error metrics, calibration, and computational cost. In contrast, replacing the weighted adjacency matrix with a binary one produces a clearer degradation in forecasting accuracy, particularly at longer horizons. This suggests that the original edge weights provide useful information beyond graph connectivity alone.

The probabilistic analysis reveals a different aspect of model performance. The Graph-Temporal model produces sharper intervals and lower quantile losses than the Temporal model, but its empirical coverage is slightly below the nominal 80% level. More importantly, this reliability degradation becomes stronger under distribution shift. During high-congestion periods, both point forecasting errors and prediction interval widths increase, while coverage decreases at longer horizons. The model therefore recognizes part of the increased uncertainty associated with congestion, but does not fully capture the shifted predictive distribution.

The mitigation experiments also reveal that calibration and robustness can be addressed at different stages of the forecasting pipeline. CQR effectively corrects the general undercoverage of the model and brings marginal coverage close to the nominal 80% level. Congestion-aware training instead targets the difficult regime directly during optimization. With $\lambda = 1$, it reduces high-congestion forecasting errors at all evaluated horizons, with the largest MAE reduction occurring at 60 minutes, while also improving coverage and producing narrower intervals than the original model under high congestion. This improvement is accompanied by a moderate degradation in normal-condition point accuracy, highlighting a trade-off between performance in the dominant regime and robustness to rare congestion events.

Nevertheless, neither mitigation eliminates the deterioration associated with longer forecasting horizons. Point forecasting errors increase with the forecast horizon, particularly under high-congestion conditions; CQR leaves these errors unchanged by construction, while congestion-aware training reduces them without eliminating their horizon-dependent increase. Empirical coverage also becomes more difficult to maintain at longer horizons, especially under high congestion. At the same time, prediction intervals remain substantially wider than under normal conditions, reducing their sharpness and informativeness.

10.2 Limitations

Several limitations should be considered when interpreting the results. First, the experiments are conducted exclusively on METR-LA and therefore reflect the spatial structure, traffic dynamics, and missing-data patterns of a single road network. The conclusions cannot necessarily be generalized to other cities or sensor networks without additional evaluation.

Second, the definition of distribution shift considered in this work focuses specifically on high-congestion periods. Although this represents a relevant and challenging traffic regime, other forms

of shift may affect forecasting models differently, including sensor failures, changes in network topology, or temporal changes in traffic patterns.

Third, the Graph-Temporal architecture is intentionally compact and explores only a limited set of spatial configurations. The diffusion-depth and adjacency ablations provide evidence about the role of graph information, but they do not constitute an exhaustive exploration of graph neural network architectures or graph construction strategies.

Finally, CQR is calibrated globally for each forecasting horizon using the validation distribution. While this substantially improves marginal coverage, it does not explicitly account for traffic regime and therefore cannot guarantee the same calibration quality within the high-congestion subset.

The congestion-aware mitigation also has limitations. The weighting strategy depends on a binary definition of high congestion and on the choice of the reweighting parameter λ . The evaluated $\lambda = 1$ configuration improves performance in the targeted regime but slightly degrades point accuracy under normal conditions. A broader validation-based analysis of the weighting strength would therefore be required to characterize this trade-off more completely.

10.3 Future Improvements

Several extensions could address these limitations. A first extension would be to investigate the congestion-aware training strategy more systematically by validating different values of the weighting parameter λ . This would allow the trade-off between robustness under high congestion and performance under normal traffic conditions to be characterized more precisely. More adaptive weighting schemes could also be considered, for example by assigning continuous weights based on congestion severity rather than using a binary normal/high-congestion distinction.

The robustness analysis could also be extended to additional distribution shifts, such as partially masked sensor histories or simulated sensor failures. This would provide a broader characterization of the dependence of the Graph-Temporal model on spatial information and of its behavior when parts of the network become unavailable.

Finally, future work could explore more expressive spatial and temporal architectures, alternative graph construction strategies, and dynamic graphs whose relationships evolve with traffic conditions. Evaluation on additional traffic datasets would also be necessary to determine whether the conclusions observed on METR-LA generalize to different road networks and traffic regimes.

11 Conclusion

This work investigated probabilistic graph-based traffic forecasting under distribution shift using the METR-LA dataset. A leakage-free experimental pipeline was developed to compare a persistence baseline, a purely temporal GRU, and a Graph-Temporal model combining diffusion-based message passing with recurrent temporal modeling. In addition to point forecasting accuracy, the analysis considered predictive uncertainty through quantile regression, empirical coverage, and prediction interval width.

The experimental results show that incorporating the road-network graph consistently improves forecasting accuracy over the Temporal model, particularly at longer forecasting horizons. The graph ablations further indicate that the weighted connectivity structure contains useful predictive information, while increasing the diffusion depth provides only limited additional benefits. These improvements, however, come with higher computational cost and slightly reduced empirical coverage.

The high-congestion stress test reveals a more challenging limitation. During congested periods, forecasting errors increase substantially and the model responds by producing wider prediction intervals. Nevertheless, coverage deteriorates at longer horizons, showing that increased uncertainty does not necessarily translate into reliable uncertainty estimates under distribution shift.

Two complementary mitigation strategies were evaluated. Post-hoc calibration through Conformalized Quantile Regression substantially improves overall coverage and brings it close to the nominal 80% level without modifying the median forecasts or retraining the model. However, residual undercoverage remains under severe high congestion, particularly at the longest forecasting horizon.

Congestion-aware training addresses this limitation from a different perspective by assigning greater importance to high-congestion targets during optimization. The implemented weighted Pinball Loss improves point forecasting accuracy under congestion at all evaluated horizons, with the largest gains at 60 minutes. It also increases empirical coverage while producing narrower prediction intervals in the shifted regime, indicating that the improvement does not result from simply

increasing predictive uncertainty. These gains are accompanied by a moderate degradation in point accuracy under normal conditions, revealing a trade-off between average-regime performance and robustness to rare congestion events.

Overall, the results demonstrate the value of exploiting spatial information for traffic forecasting while also highlighting the importance of evaluating predictive uncertainty beyond average test performance. Reliable deployment requires considering not only forecasting accuracy, but also calibration, robustness to distribution shift, and the computational cost associated with more expressive spatio-temporal models.

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